How the Web Works

Chapter 1

Slides adapted from

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Fundamentals of Web Development

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Objectives

1. Definitions and History
2. Internet Protocols
3. Client-Server Model
4. Where is the Internet?
5. Domain Name System
6. Uniform Resource Locators (URL)
7. Hypertext Transfer Protocol (HTTP)
8. Web Servers

Section 1 of 8

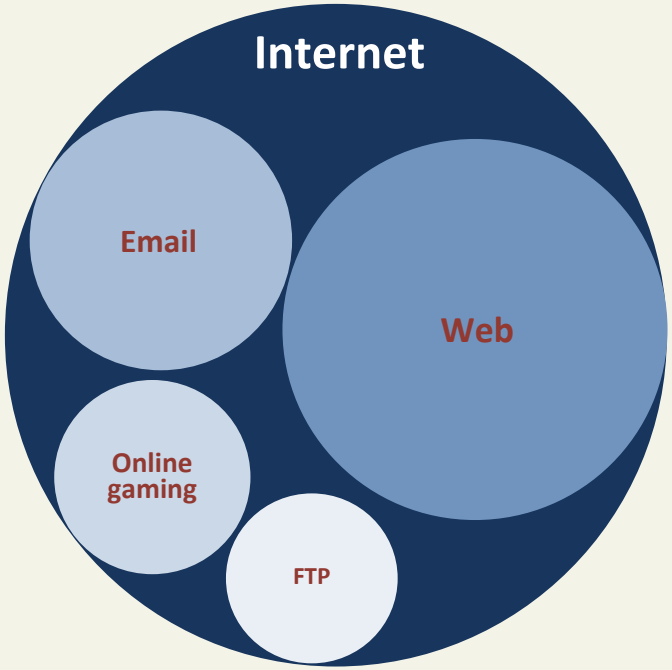
DEFINITIONS AND HISTORY

Internet = Web?

The answer is no

The World-Wide Web (WWW or simply the Web) is certainly what most people think of when they see the word “Internet.”

But the WWW is only a part of the Internet.



Communication Definitions

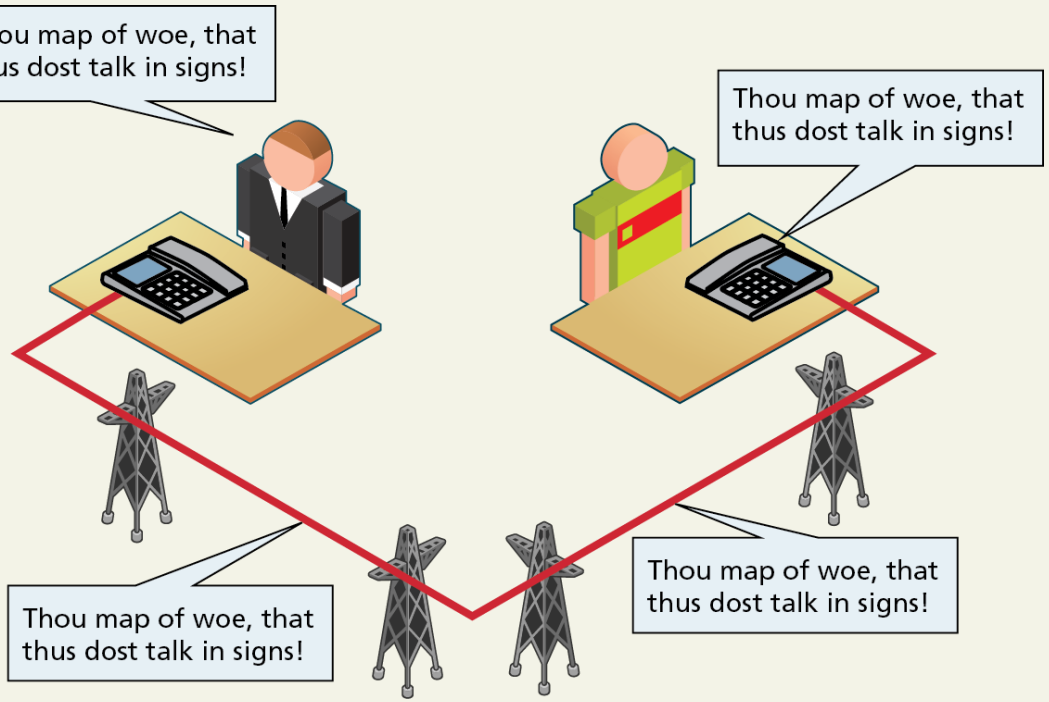
We will begin with the telephone

Telephone networks provide a good starting place to learn about modern digital communications.

In the telephone networks of old, calls were routed through operators who physically connected caller and receiver by connecting a wire to a switchboard to complete the circuit.

Circuit Switching

A **circuit switching** establishes an actual physical connection between two people through a series of physical switches.



Circuit Switching

Its Limitations

Circuit Switching Weaknesses

- You must establish a link and maintain a dedicated circuit for the duration of the call
- Difficult to have multiple conversations simultaneously
- Wastes bandwidth since even the silences are transmitted

ARPANET

The beginnings of the Internet

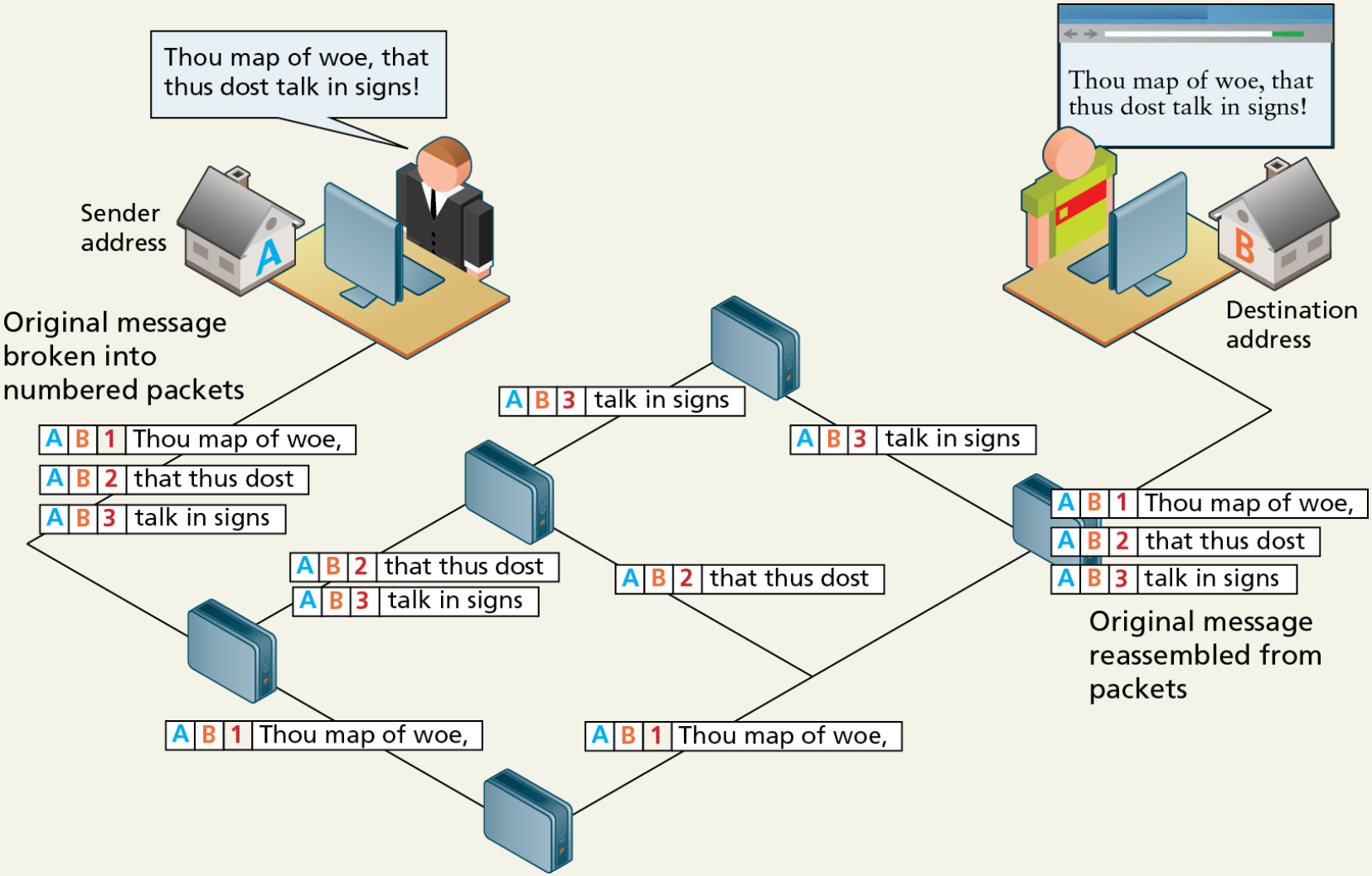
The research network ARPANET was created. In the 1960s

- ARPANET did not use circuit switching
- it used **packet switching**

A packet-switched network does not require a continuous connection. Instead, it splits the messages into smaller chunks called **packets** and routes them to the appropriate place based on the destination address.

The **packets** can take different routes to the destination.

Packet Switching



Packet Switching

Isn't this more complicated?

While **packet switching** may seem a more complicated and inefficient approach than **circuit switching**, it is:

- more robust (it is not reliant on a single pathway that may fail) and
- a more efficient use of network resources (since a circuit can communicate multiple connections).

Short History of the Internet

Perhaps not short enough

The early ARPANET network was funded and controlled by the United States government, and was used exclusively for academic and scientific purposes.

The early network started small with just a handful of connected campuses in 1969 and grew to a few hundred by the early 1980s.

TCP/IP

Rides to the rescue

To promote the growth and unification of the disparate networks a suite of protocols was invented to unify the networks together.

By 1981, new networks built in the US began to adopt the **TCP/IP** (**Transmission Control Protocol / Internet Protocol**) communication model (discussed in the next section), while older networks were transitioned over to it.

Tim Berners-Lee

I meant Sir Tim Berners-Lee

The invention of the WWW is usually attributed to the British Tim Berners-Lee, who, along with the Belgian Robert Cailliau, published a proposal in 1990 for a hypertext system while both were working at CERN in Switzerland.

Core Features of the Web

Shortly after that initial proposal Berners-Lee developed the main features of the web:

1. A URL to uniquely identify and locate a resource on the WWW.
2. The HTTP protocol to describe how requests and responses operate.
3. HTML to publish documents.
4. A software program (later called web server software) that can respond to HTTP requests.
5. A program (later called a browser) to make HTTP requests from URLs and that can display the HTML it receives.

W3C

The World Wide Web Consortium

Also in late 1994, Berners-Lee helped found the **World Wide Web Consortium (W3C)**, which would soon become the international standards organization that would oversee the growth of the web.

This growth was very much facilitated by the decision of CERN to not patent the work and ideas done by its employee and instead left the web protocols and code-base royalty free.

Web Apps Compared to Desktop Apps

First the advantages of web apps

Some of the **advantages** of web applications include:

- Accessible from any Internet-enabled computer.
- Usable with different operating systems and browser platforms.
- Easier to roll out program updates since only need to update software on server and not on every desktop in organization.
- Centralized storage on the server means fewer concerns about local storage (which is important for sensitive information such as health care data).

Web Apps Compared to Desktop Apps

Now the disadvantages of web apps

Some of the **disadvantages** of web applications include:

- Requirement to have an active Internet connection (the Internet is not always available everywhere at all times).
- Security concerns about sensitive private data being transmitted over the Internet.
- Concerns over the storage, licensing and use of uploaded data.
- Problems with certain websites on certain browsers not looking quite right.
- Limited access to the operating system can prevent software and hardware from being installed or accessed (like Adobe Flash on iOS).

What is an "Intranet"?

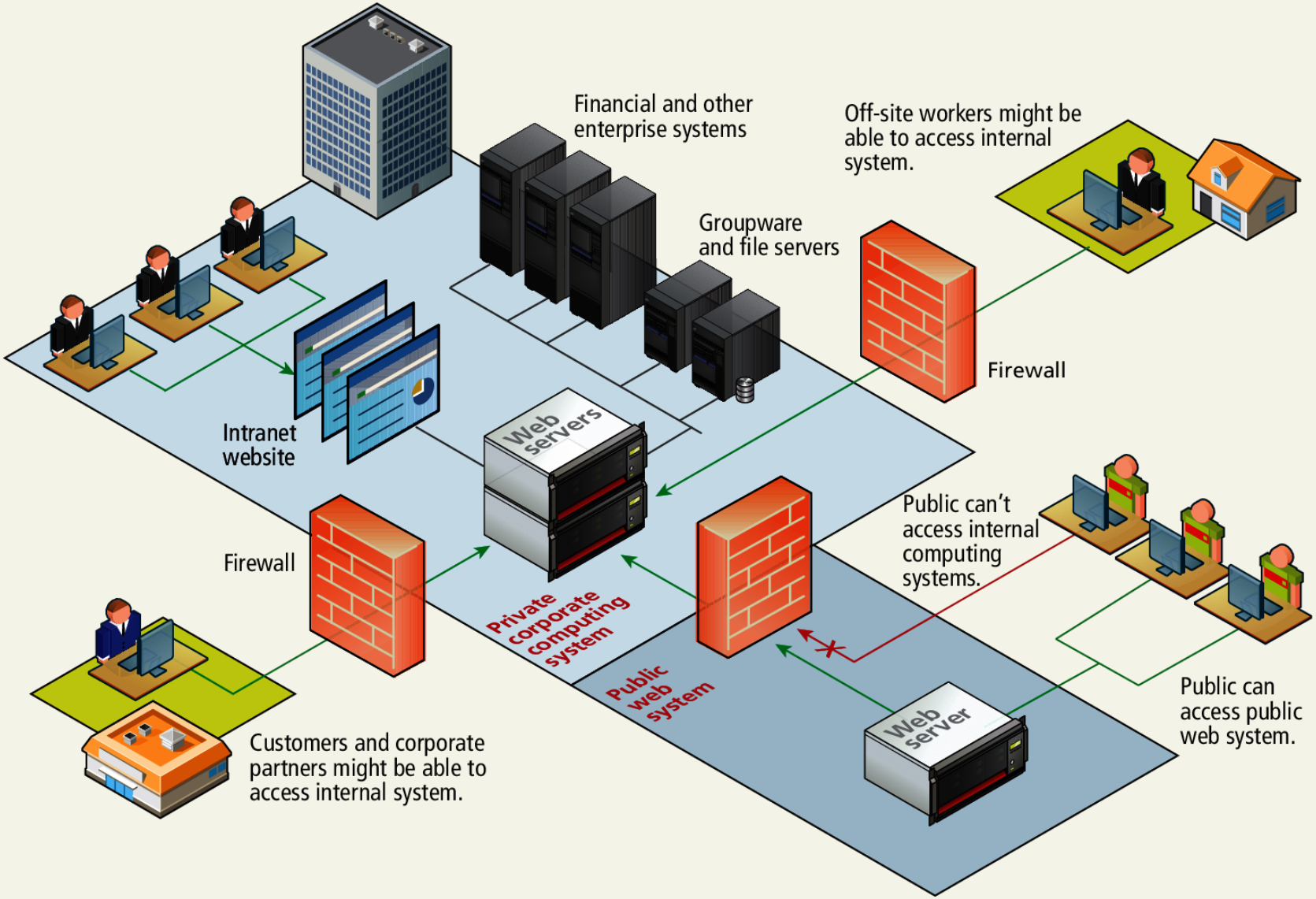
A short digression

A common term you might encounter in web development is the term "**intranet**" (with an "**a**"), which refers to an internet network that is local to an organization or business.

Intranet resources are often private, meaning that only employees (or authorized external parties such as customers or suppliers) have access to those resources.

Thus Internet (with an "e") is a broader term that encompasses both private (intranet) and public networked resources.

Intranet versus Internet



Static Web Sites

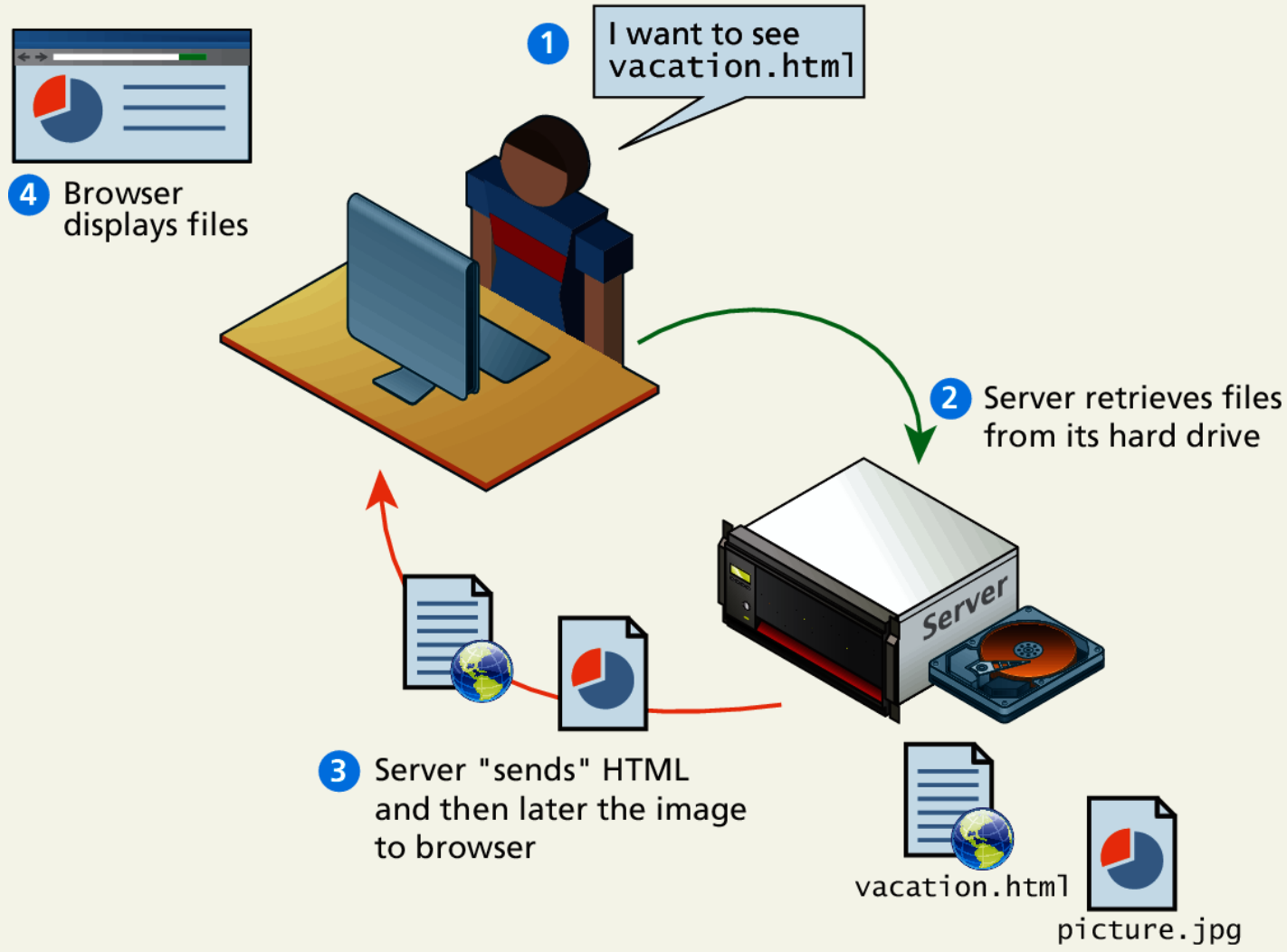
Partying Like It's 1995

In the earliest days of the web, a **webmaster** (the term popular in the 1990s for the person who was responsible for creating and supporting a web site) would publish web pages, and periodically update them.

In those early days, the skills needed to create a web site were pretty basic: one needed knowledge of the HTML markup language and perhaps familiarity with editing and creating images.

This type of web site is commonly referred to as a **static web site**, in that it consists only of HTML pages that look identical for all users at all times.

Static Web Sites



Dynamic Web Sites

What are they?

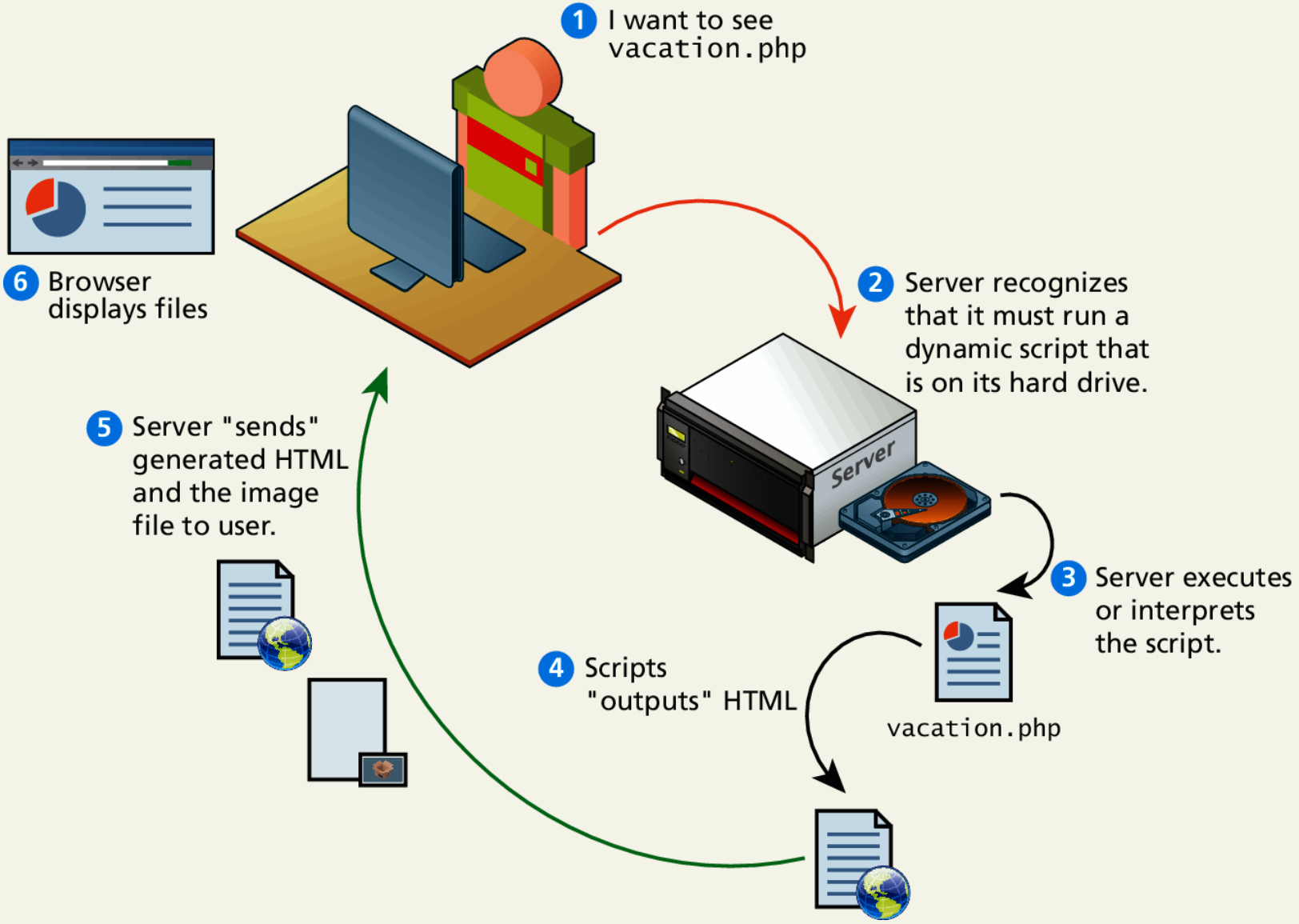
Within a few years of the invention of the web, sites began to get more complicated as more and more sites began to use **programs** running on web servers to generate content **dynamically**.

These server-based programs would read content from databases, interface with existing enterprise computer systems, communicate with financial institutions, and then output HTML that would be sent back to the users' browsers.

This type of web site is called here in this book a **dynamic web site** because the page content is being created at run-time by a program created by a programmer.

Page content can vary for user to user.

Dynamic Web Sites



Web 2.0 and Beyond

In the mid 2000s, a new buzz-word entered the computer lexicon: **web 2.0**.

This term had two meanings, one for users and one for developers.

- For the users, Web 2.0 referred to an interactive experience where users could contribute *and* consume web content, thus creating a more user-driven web experience.
- For software developers, Web 2.0 also referred to a change in the paradigm of how dynamic web sites are created.
  - Programming logic, which previously existed only on the server, began to migrate to the browser.
  - This required learning **Javascript**, a rather tricky programming language that runs in the browser, as well as mastering the rather difficult programming techniques involved in asynchronous communication.

Section 2 of 8

INTERNET PROTOCOLS

What's a Protocol?

The Internet exists today because of a suite of interrelated communications protocols.

A **protocol** is a set of rules that partners in communication use when they communicate.

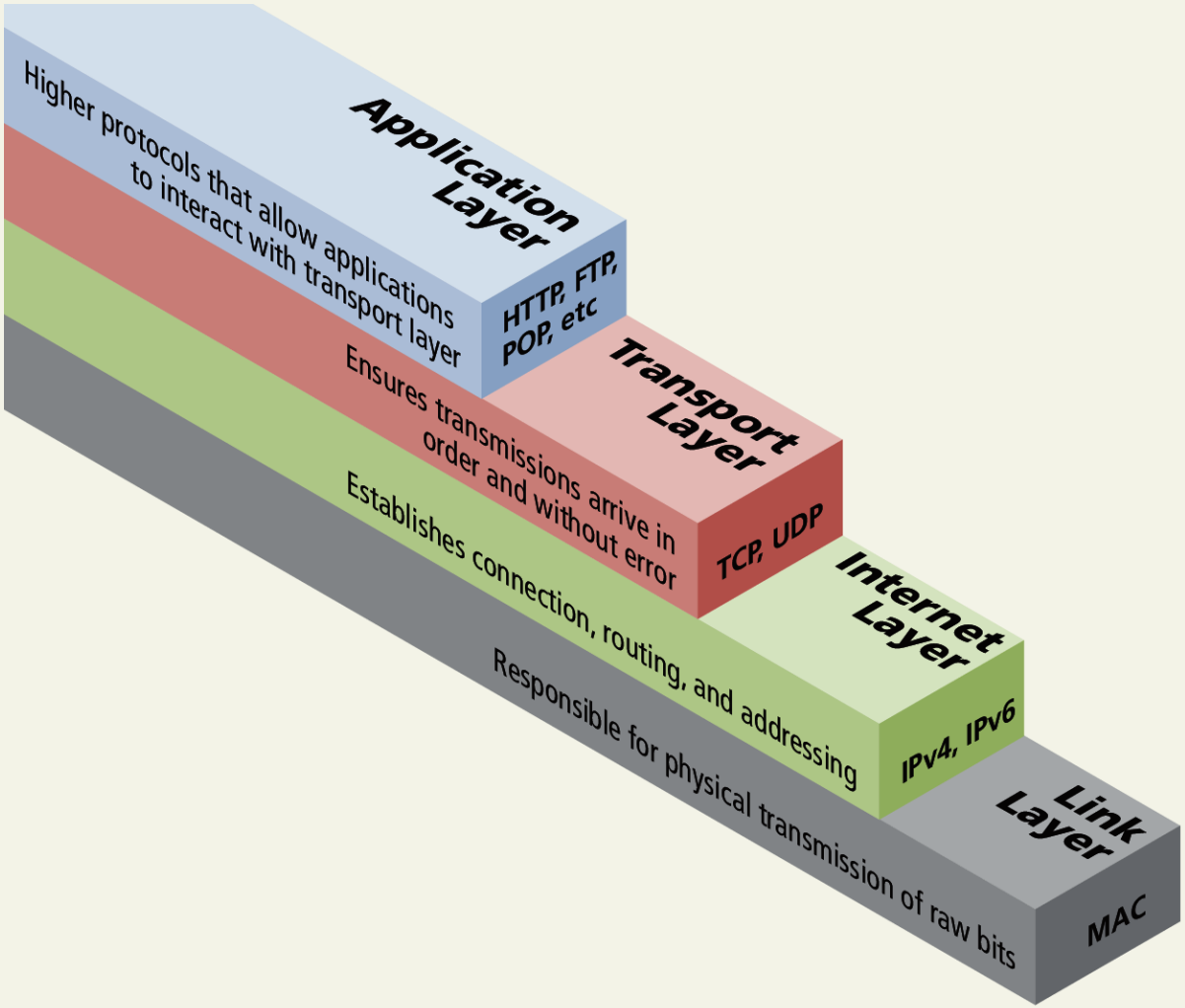
A Layered Architecture

The TCP/IP Internet protocols were originally abstracted as a four-layer stack.

Later abstractions subdivide it further into five or seven layers.

Since we are focused on the top layer anyhow, we will use the earliest and simplest **four-layer network model**.

Four Layer Network Model



Link Layer

Save this for your networking course

- The **link layer** is the lowest layer, responsible for both the physical transmission across media (wires, wireless) and establishing logical links.
- It handles issues like packet creation, transmission, reception and error detection, collisions, line sharing and more.

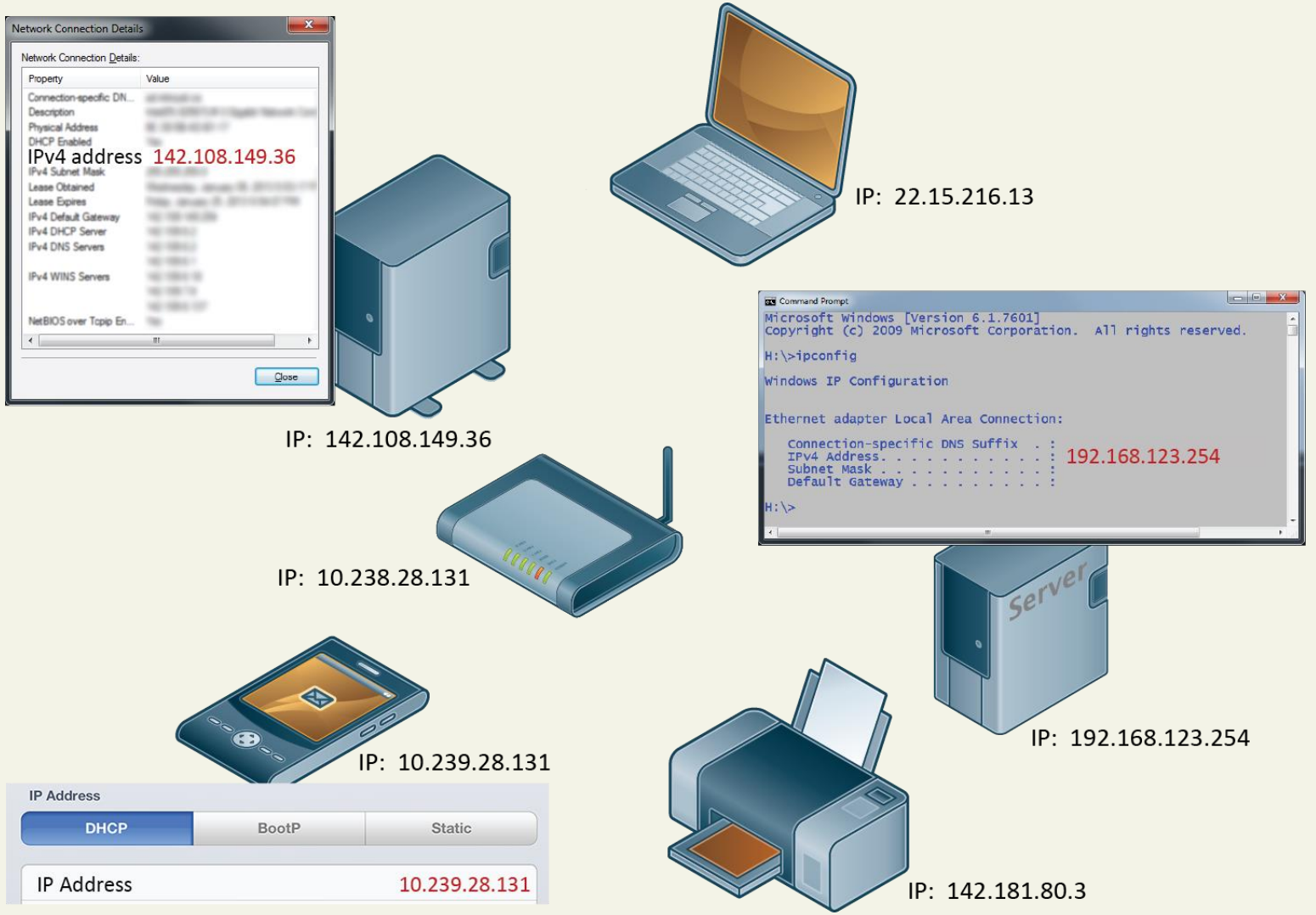
Internet Layer

The **Internet layer** (sometimes also called the IP Layer) routes packets between communication partners across networks.

Internet Protocol (IP)

- The Internet uses the **Internet Protocol** (**IP**) addresses to identify destinations on the Internet.
- Every device connected to the Internet has an **IP address**, which is a numeric code that is meant to uniquely identify it.

IP addresses and the Internet



IP Addresses

Two types

- **IPv4** addresses are the IP addresses from the original TCP/IP protocol.
- 32 bits.
- In human-readable form, IPv4 addresses are written as a sequence of 4 integers (0-255 range) separated by dots
- An unsigned 8-bit integer's maximum value is 255
- 32 bits (four integers together) can encode approximately 4.2 billion unique IP addresses.

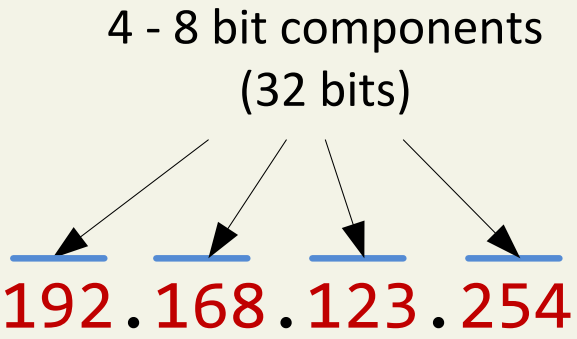
IP Addresses

Two types

- To future proof the Internet against the 4.2 billion limit, a new version of the IP protocol was created, **IPv6**.
- This newer version uses 128 bits for 2^{128} unique addresses, over a billion billion times the number in IPv4.
- In human-readable form, IPv6 addresses are written as eight 16-bit integers. These 16-bit integers are normally expressed in hexadecimal, due to their longer length.

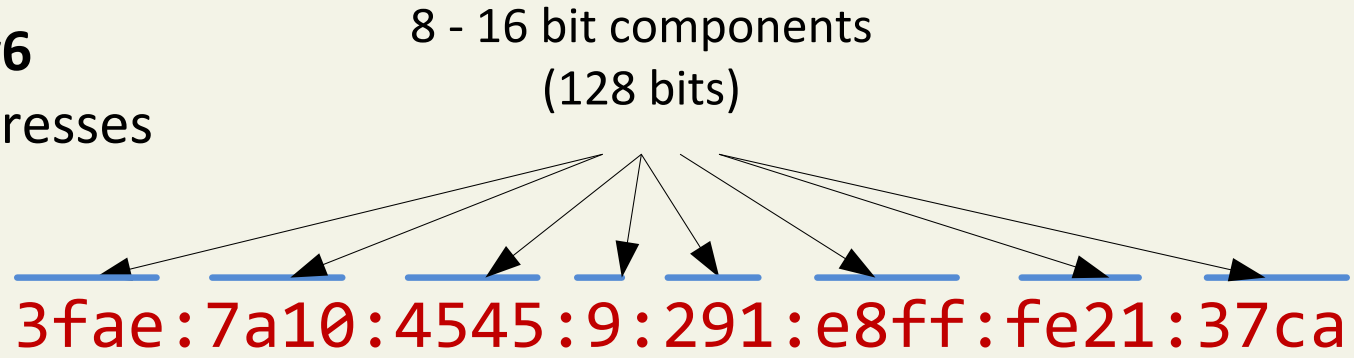
IPv4

2^{32} addresses



IPv6

2^{128} addresses



IP Addresses

Inside of networks is different

Your IP address will generally be assigned to you by your Internet Service Provider (ISP).

In organizations, large and small, purchasing extra IP addresses from the ISP is not cost effective.

In a local network, computers can share a single IP address between them.

Network Address Translation (NAT)

Transport Layer

Limits of IP (network layer):

- delivery is not guaranteed
- packets may arrive out of order
- communication is between hosts
- no flow, congestion control

The **transport layer** ensures transmissions arrive all, in order, and without error.

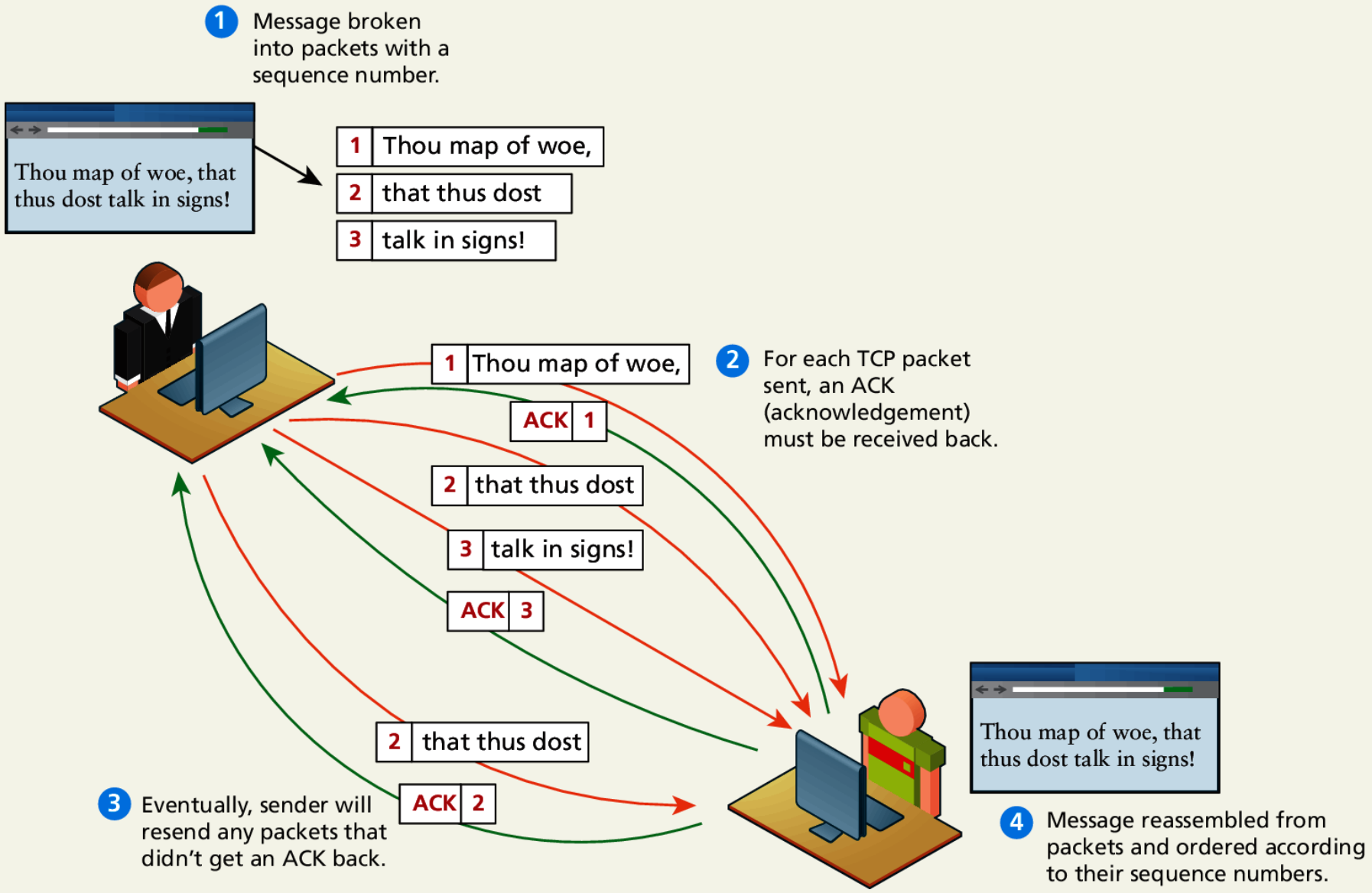
This is accomplished through a few mechanisms.

First, the data is broken into packets formatted according to the **Transmission Control Protocol** (**TCP**).

Transport Layer

- Secondly, each packet is acknowledged back to the sender so in the event of a lost packet, the transmitter will realize a packet has been lost since no ACK arrived for that packet.
- That packet is retransmitted, and although out of order, is reordered at the destination.
- Flow control and congestion control.
- Transport layer: communication is now between processes and not hosts.

TCP Packets


1 Message broken into packets with a sequence number.
Thou map of woe, that thus dost talk in signs!
1 Thou map of woe,
2 that thus dost
3 talk in signs!
1 Thou map of woe,
ACK 1
2 For each TCP packet sent, an ACK (acknowledgement) must be received back.
2 that thus dost
3 talk in signs!
ACK 3
2 that thus dost
ACK 2
3 Eventually, sender will resend any packets that didn't get an ACK back.
Thou map of woe, that thus dost talk in signs!
4 Message reassembled from packets and ordered according to their sequence numbers.

Application Layer

- With the **application layer**, we are at the level of protocols familiar to most web developers.
- Application layer protocols implement process-to-process communication that are application-specific.
- At a higher level of abstraction in comparison to the low-level packet and IP addresses protocols in the layers below it.
- Examples: **HTTP**, SSH, FTP, DNS, POP, SMTP.

Section 3 of 8

CLIENT-SERVER MODEL

Client-Server Model

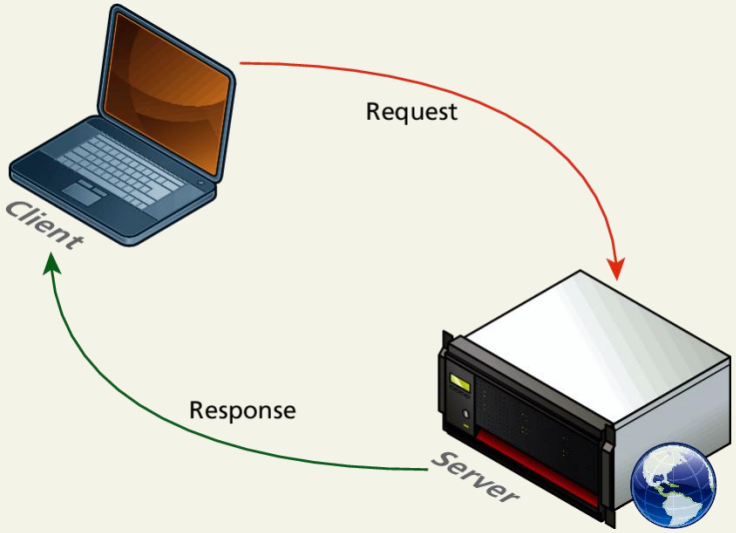
What is it?

- The web is sometimes referred to as a client-server model of communications.
- In the **client-server model**, there are two types of actors: clients and servers.
- The **server** is a computer agent that is normally active 24 hours a day, 7 days a week, listening for queries from any client who make a request.
- A **client** is a computer agent that makes requests and receives responses from the server, in the form of response codes, images, text files, and other data.

Request-Response Loop

Within the client-server model, the **request-response loop** is the most basic mechanism on the server for receiving requests and transmitting data in response.

The client initiates a **request** to a server and gets a **response** that could include some resource like an HTML file, an image or some other data.

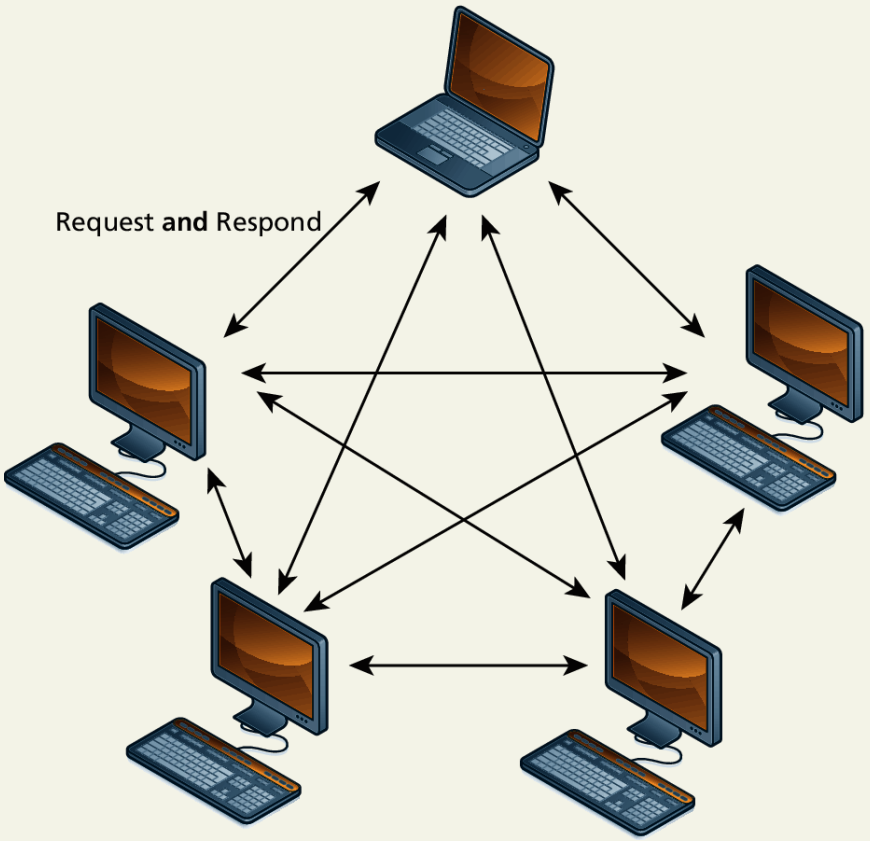


The Peer-to-Peer Alternative

Not actually illegal

In the **peer-to-peer model** where each computer is functionally identical, each node is able to send and receive directly with one another.

In such a model each peer acts as both a client and server able to upload and download information.



Server Types

A server is rarely just a single computer

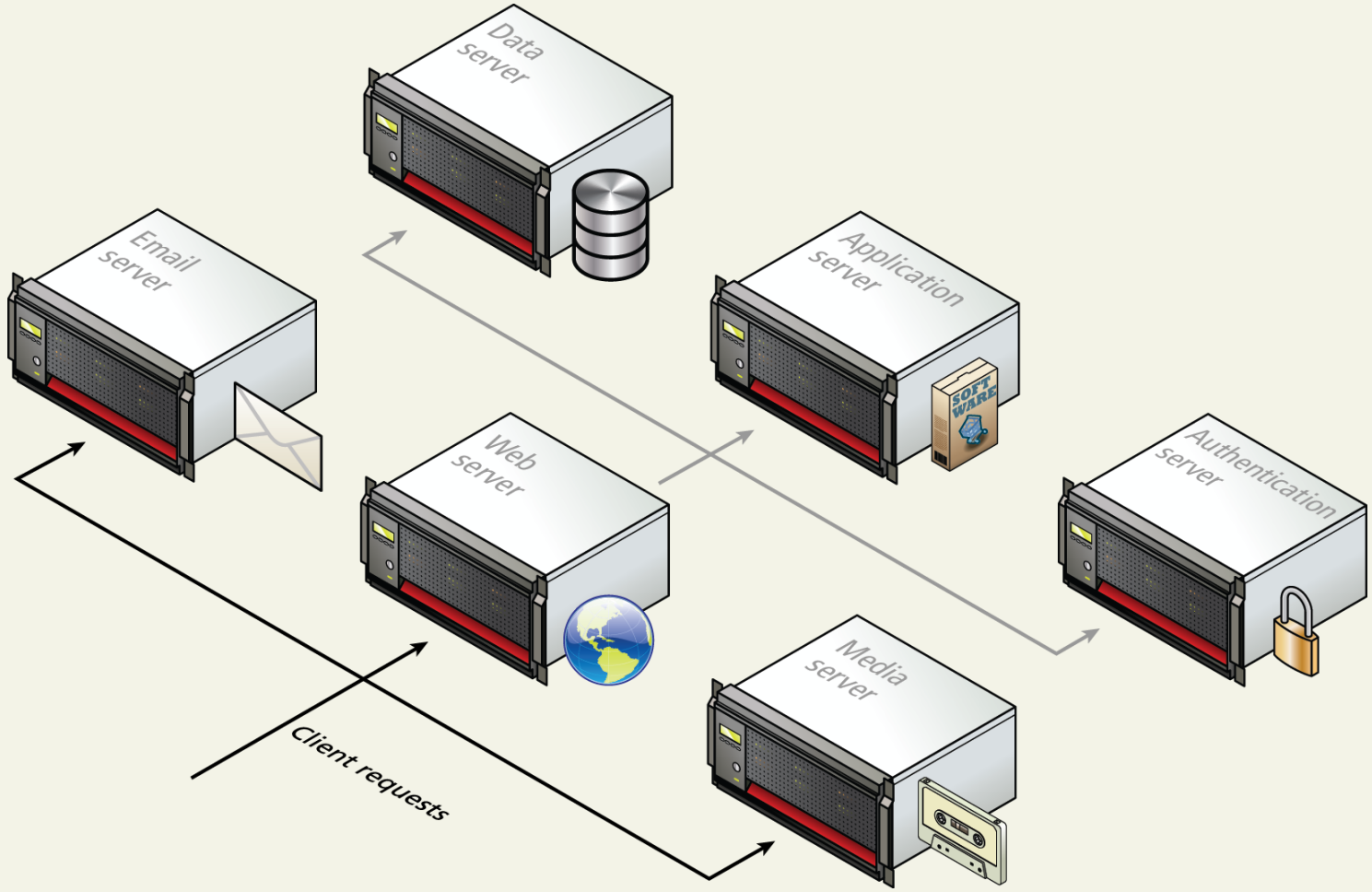
Earlier, the server was shown as a single machine, which is fine from a conceptual standpoint.

Clients make requests for resources from a URL; to the client, the server *is* a single machine.

However, most real-world web sites are typically not served from a single server machine, but by many servers.

It is common to split the functionality of a web site between several different types of server.

Server Types



Real-World Server Installations

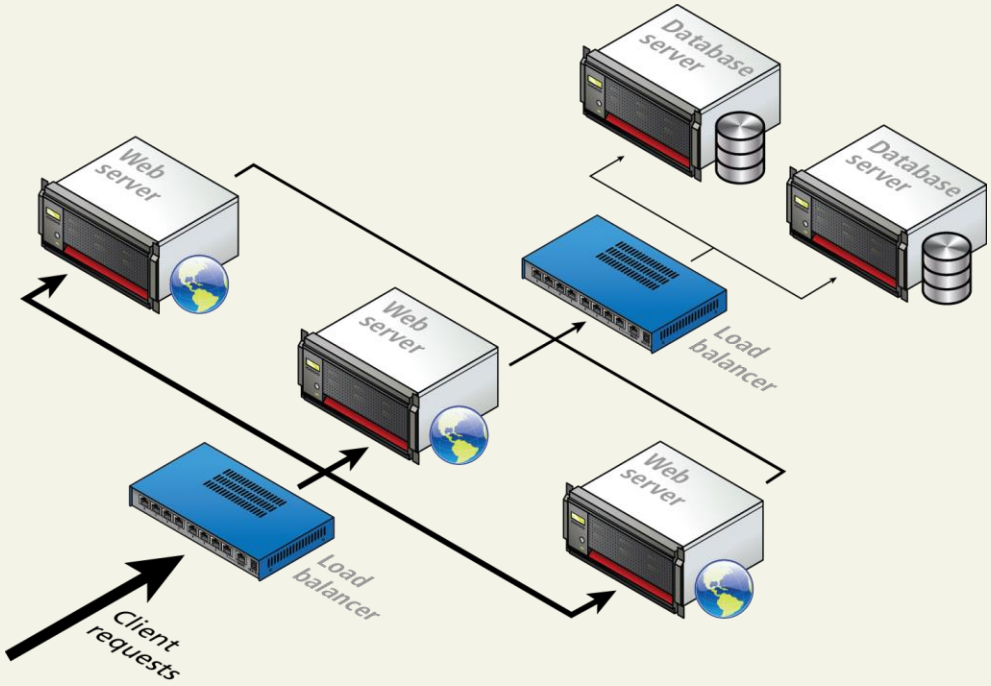
Not only are there different types of servers, there is often replication of each of the different server types.

A busy site can receive thousands or even tens of thousands of requests a second; globally popular sites such as Facebook receive millions of requests a second.

Server Farms

Have no cows

A single web server that is also acting as an application or database server will be hard-pressed to handle more than a few thousands requests a second, so the usual strategy for busier sites is to use a **server farm**.

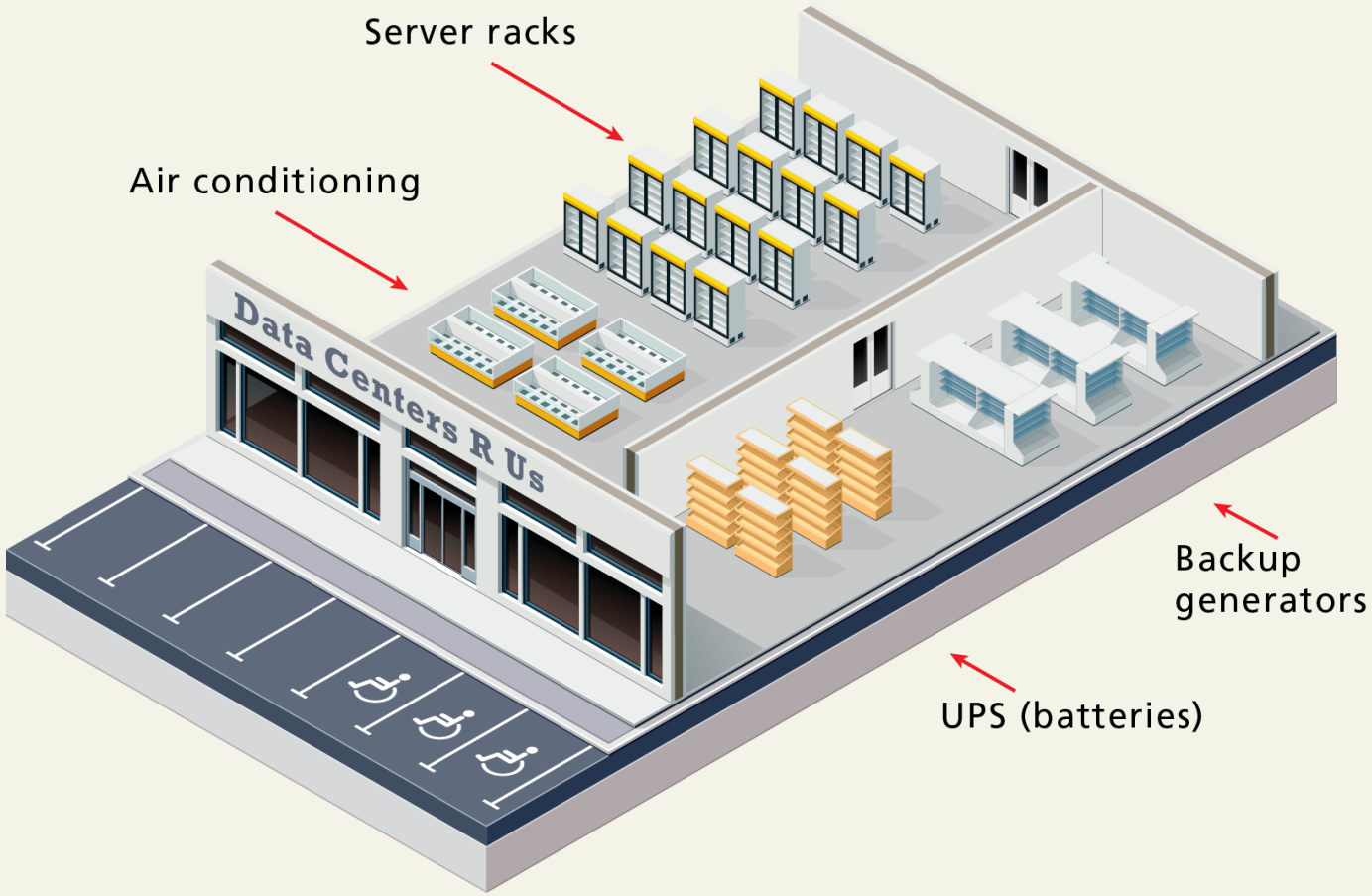


Server Farms

- The goal behind server farms is to distribute incoming requests between clusters of machines so that any given web or data server is not excessively overloaded.
- Special routers called **load balancers** distribute incoming requests to available machines.
- Even if a site can handle its load via a single server, it is not uncommon to still use a server farm because it provides **failover redundancy**.
- That is, if the hardware fails in a single server, one of the replicated servers in the farm will maintain the site's availability.

Data Center

Server farms are typically housed in special facilities called **data centers**.



Data Centers & Web Hosting

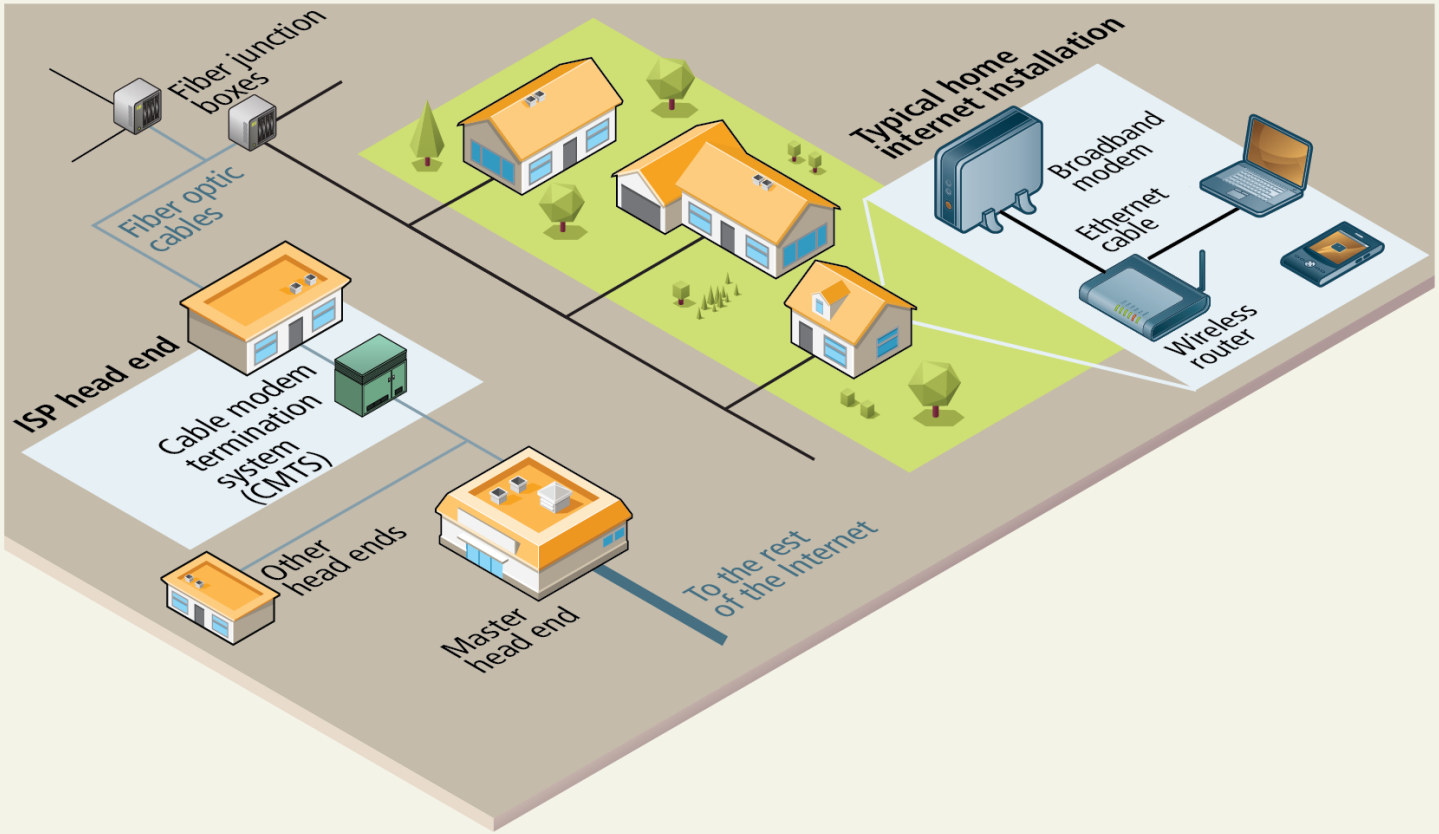
- To prevent the potential for site down times, most large web sites will exist in mirrored data centers in different parts of the country, or even world.
- As a consequence, the costs for multiple redundant data centers are quite high, and only larger web companies can afford to create and manage their own.
- Most web companies will instead lease space from a third-party data center.
- It is also common for the reverse to be true – that is, a single server machine may host multiple sites.
- Large commercial web hosting companies such as GoDaddy, Blue Host, Dreamhost, and others will typically host hundreds or even thousands of websites on a single machine (or mirrored on several servers).

Section 4 of 8

WHERE IS THE INTERNET?

From the Computer to the Local Provider

Our main experience of the hardware component of the Internet is that which we experience in our homes.



In the House

The **broadband modem** (also called a cable modem or DSL modem) is a bridge between the network hardware outside the house (typically controlled by a phone or cable company) and the network hardware inside the house.

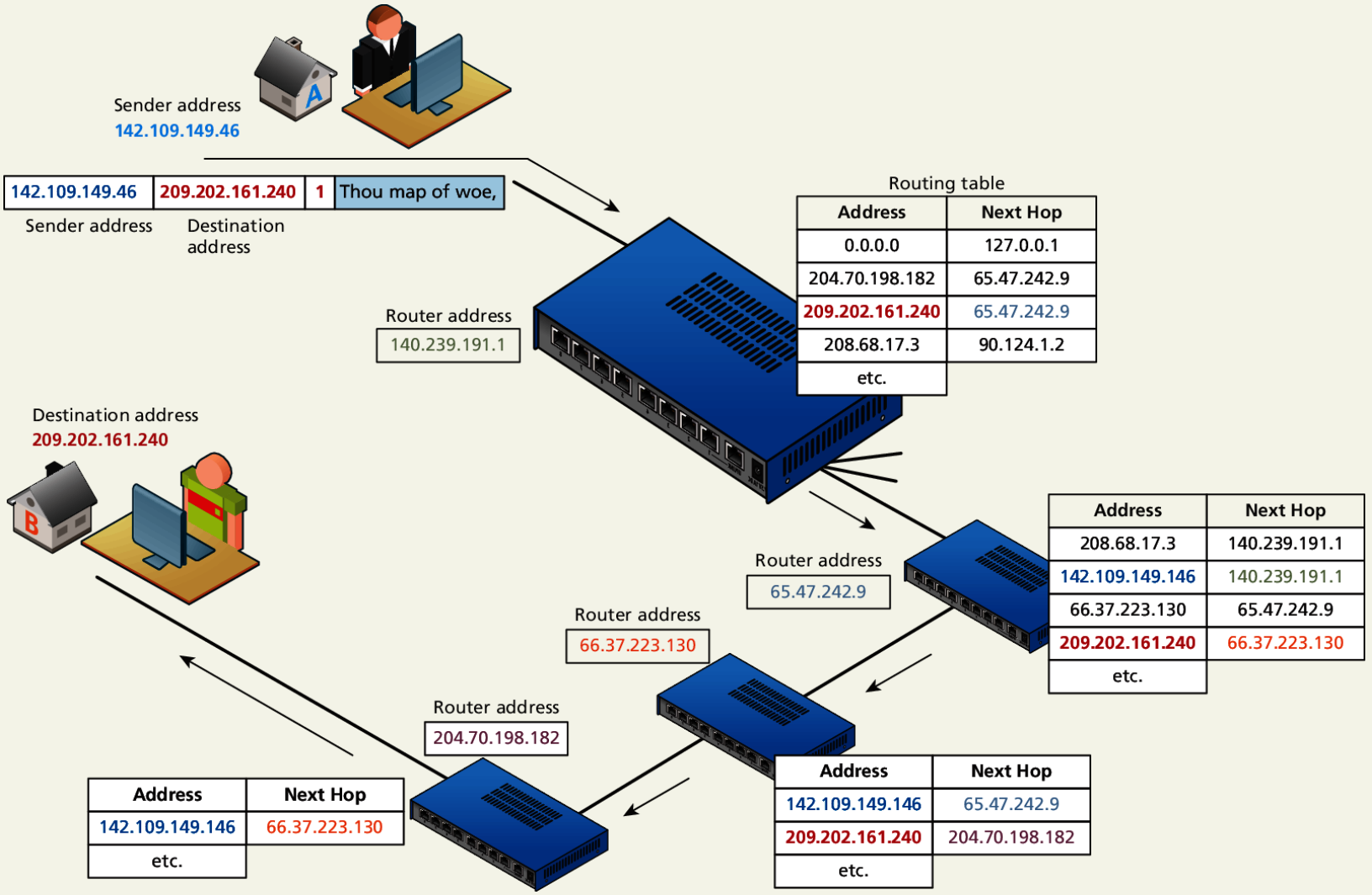
These devices are often supplied by the ISP.

The **wireless router** is perhaps the most visible manifestation of the Internet in one's home, in that it is a device we typically need to purchase and install.

Routers are in fact one of the most important and ubiquitous hardware devices that makes the Internet work.

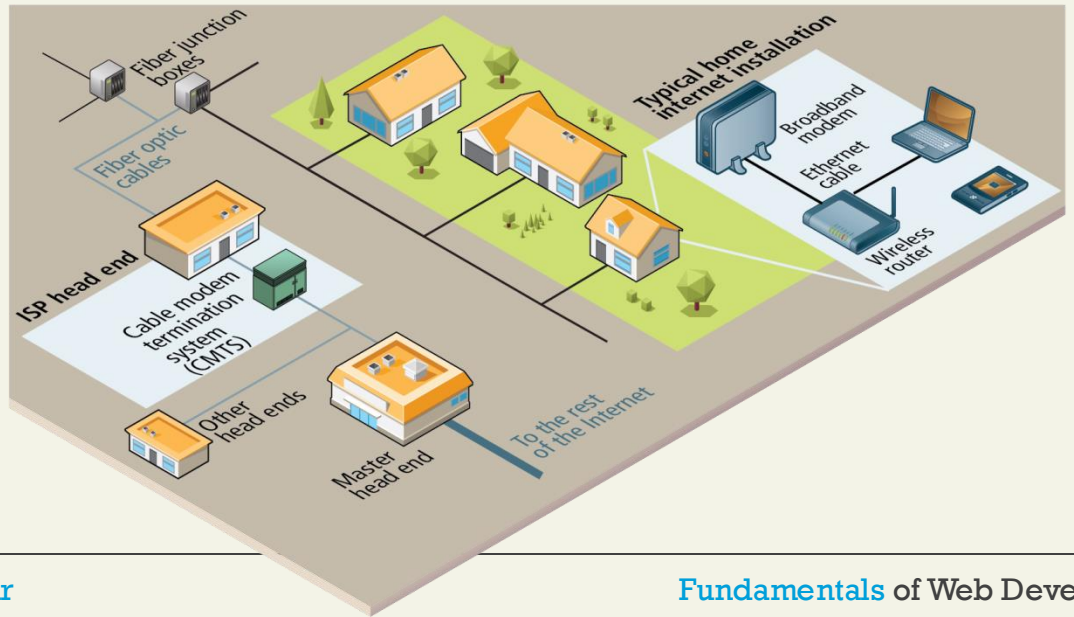
At its simplest, a **router** is a hardware device that forwards data packets from one network to another network.

Routers and Routing Tables



To the Provider

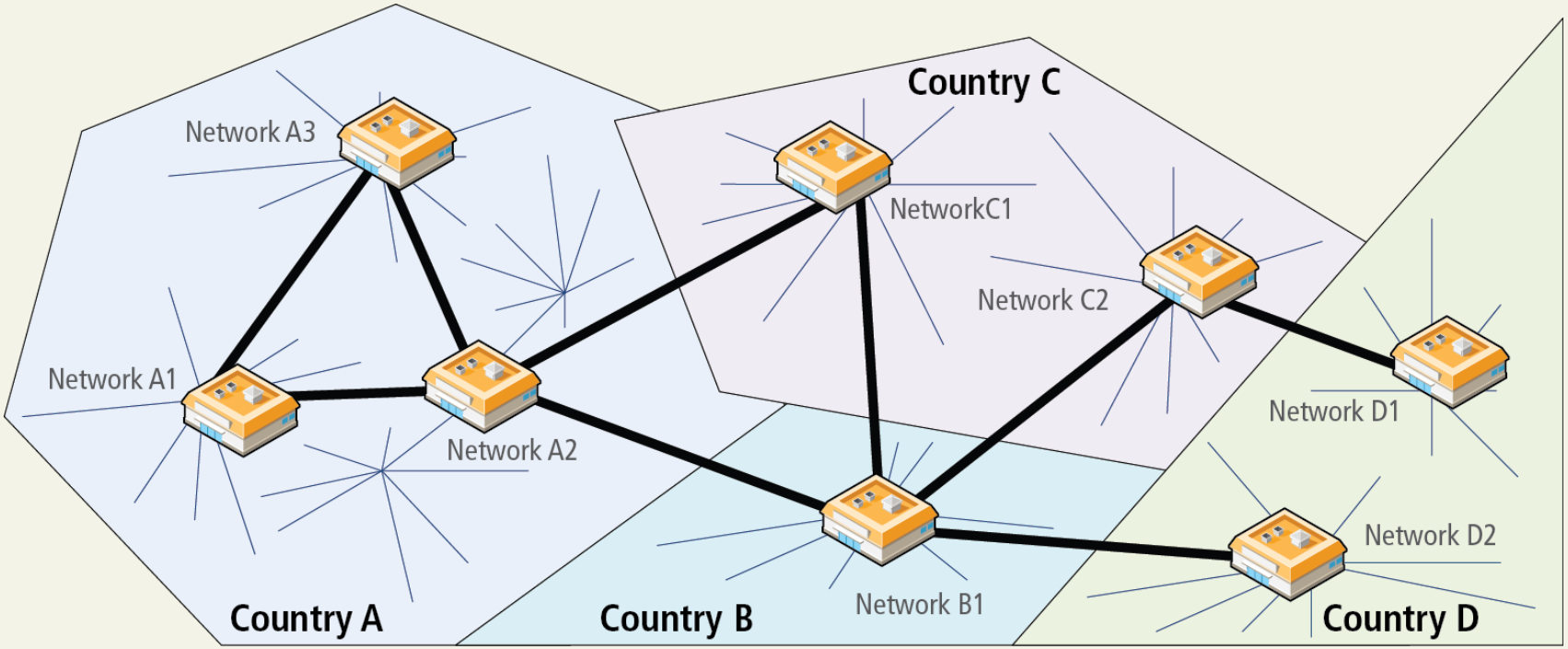
Fiber optic cables eventually make their way to an ISP's **head-end**, which is a facility that may contain a **cable modem termination system** (CMTS) or a digital subscriber line access multiplexer (DSLAM) in a DSL-based system.



From the Local Provider to the Ocean

- Eventually your ISP has to pass on your requests for Internet packets to other networks.
- This intermediate step typically involves one or more regional network hubs.
- Your ISP may have a large national network with optical fiber connecting most of the main cities in the country.
- Some countries have multiple national or regional networks, each with their own optical network.

Connecting different networks within and between countries



Internet Exchange Points

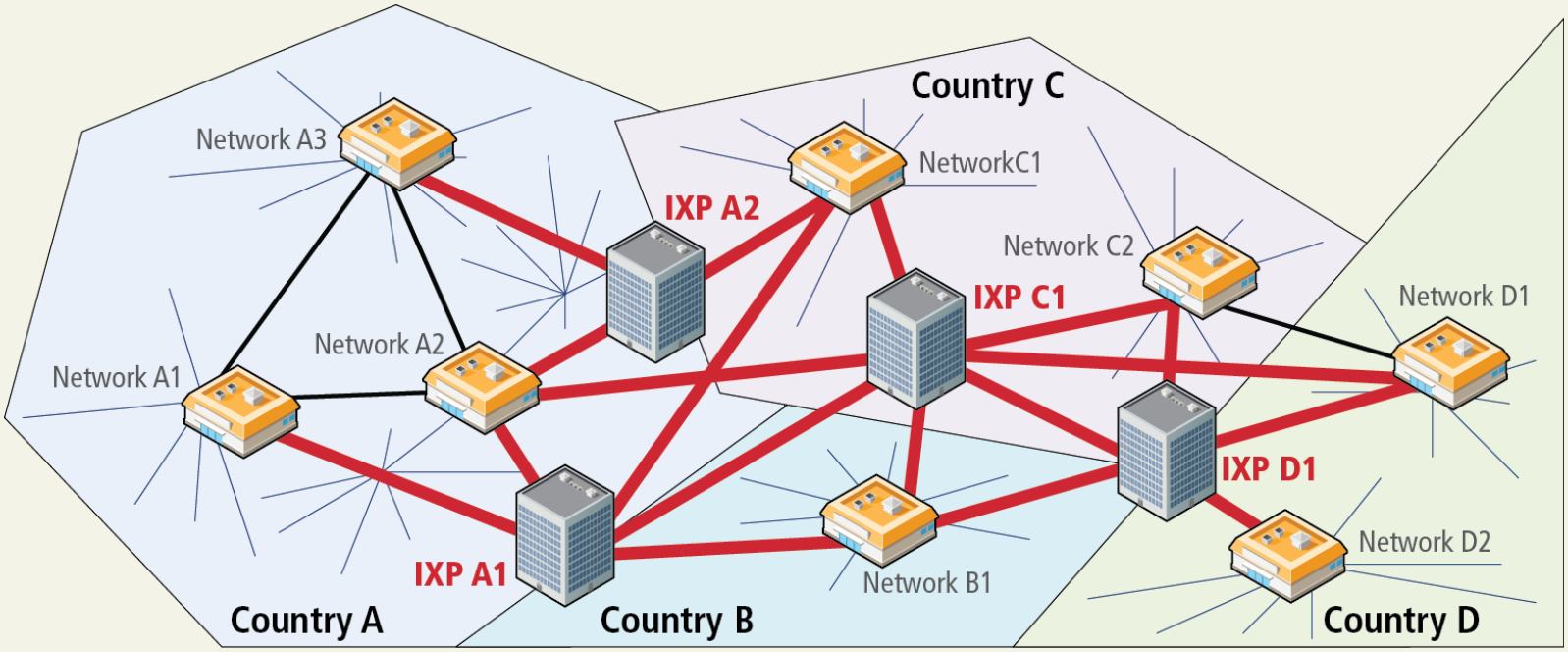
Connecting different networks

This type of network configuration began to change in the 2000s, as more and more networks began to interconnect with each other using an **Internet Exchange Point** (**IXP**).

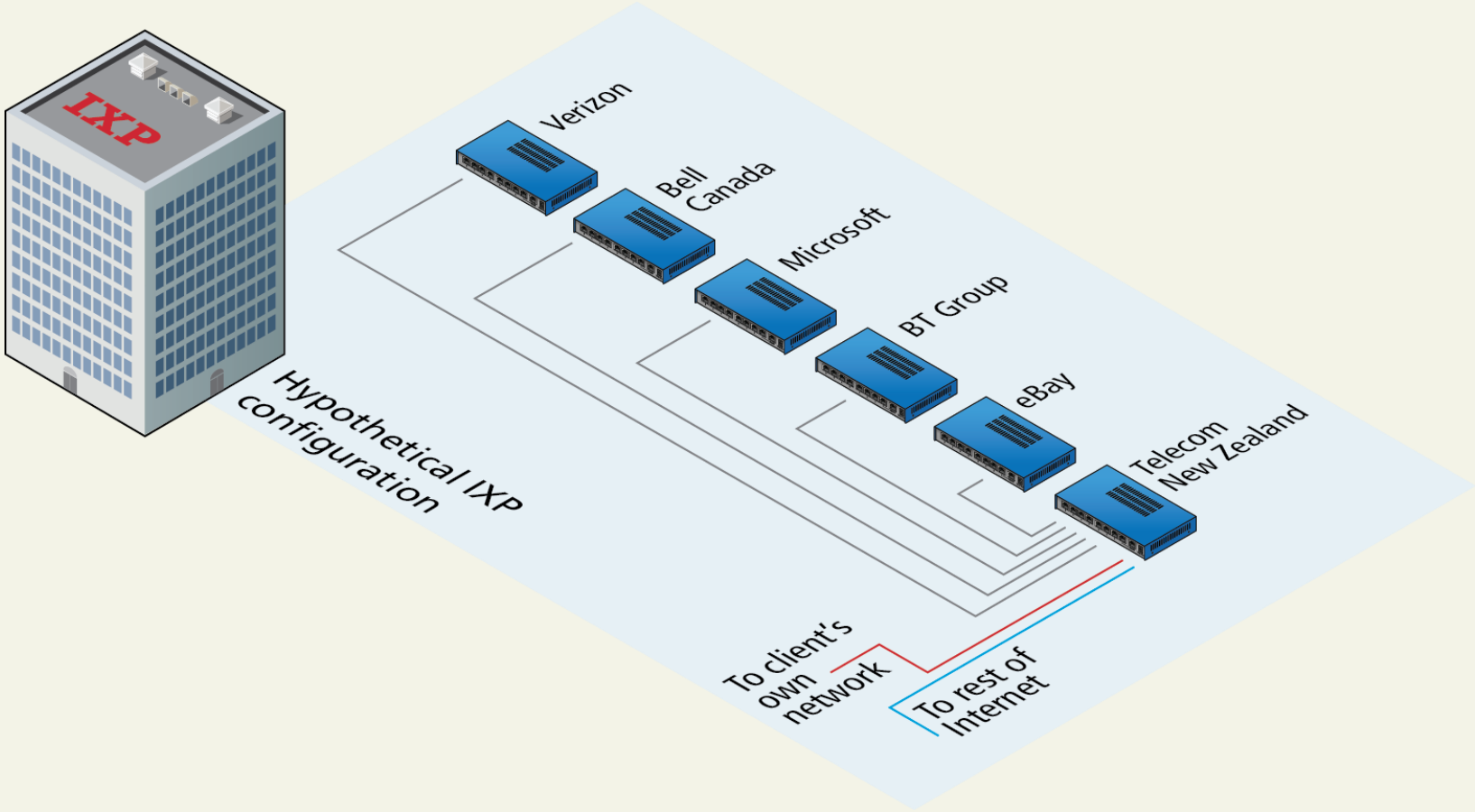
These IXPs allow different ISPs to **peer** with one another (that is, interconnect) in a shared facility, thereby improving performance for each partner in the peer relationship.

Not only IXPs, but also transit providers

National and regional networks using Internet Exchange Points



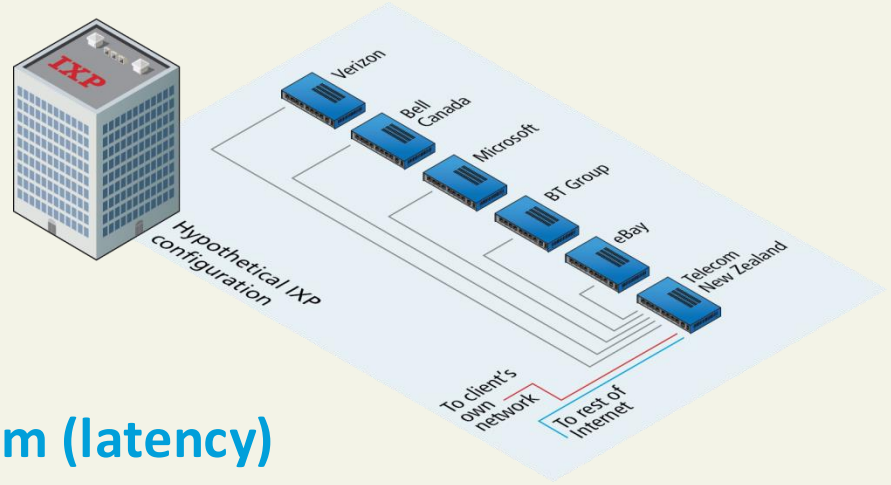
Sample Internet Exchange Point



IXPs

Not just for large networks

Different networks connect not only to other networks within an IXP, but now large web sites such as Microsoft and FaceBook are also connecting to multiple other networks simultaneously as a way of improving the performance of their sites.



Physical distance is a problem (latency)

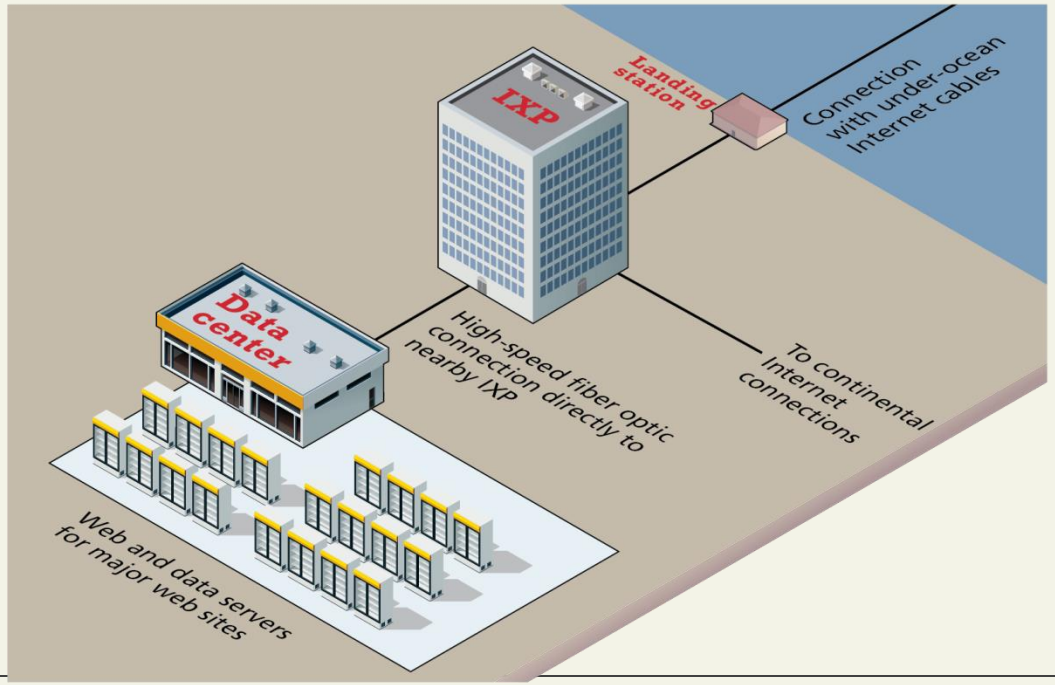
Real IXPs

- Real IXPs, such as at Palo Alto (PAIX), Amsterdam (AMS-IX), Frankfurt (DE-CIX), London (LINX), allow many hundreds of networks and companies to interconnect and have throughput in the order of terabits per second.
- The scale of peering in these IXPs is way beyond that shown in the diagram (which shows peering with only five others):
- Companies within these IXPs use large routers that have hundreds of ports allowing hundreds of simultaneous peering relationships.

IXPs and Data Centers

In recent years, major web companies have joined the network companies in making use of IXPs.

As shown in the diagram, this sometimes involves mirroring a site's infrastructure (i.e., web and data servers) in a data center located near the IXP.

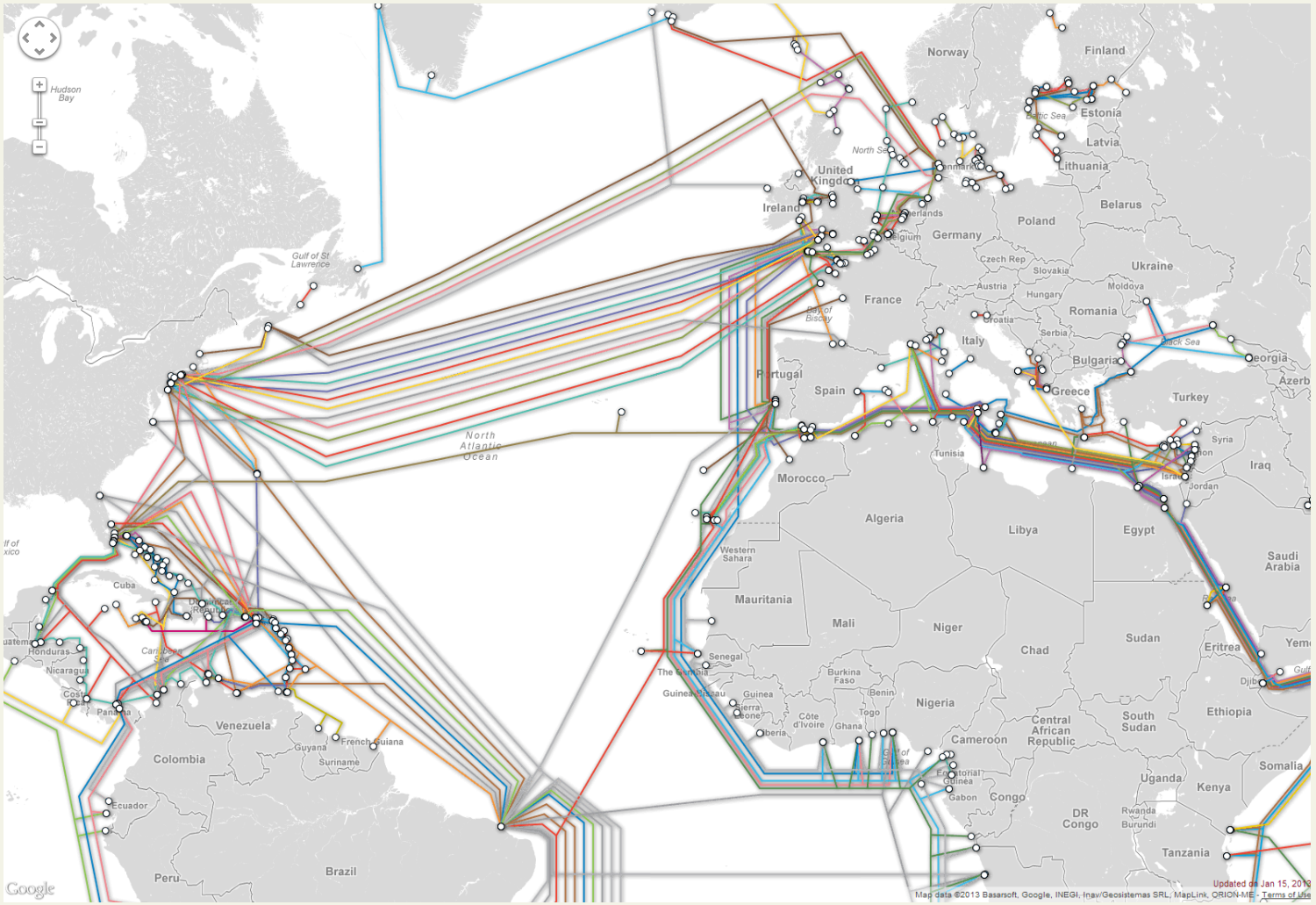


Across the Oceans

Eventually, international Internet communication will need to travel underwater.

The amount of undersea fiber optic cable is quite staggering and is growing yearly.

Undersea fiber optic lines (courtesy TeleGeography)



Section 5 of 8

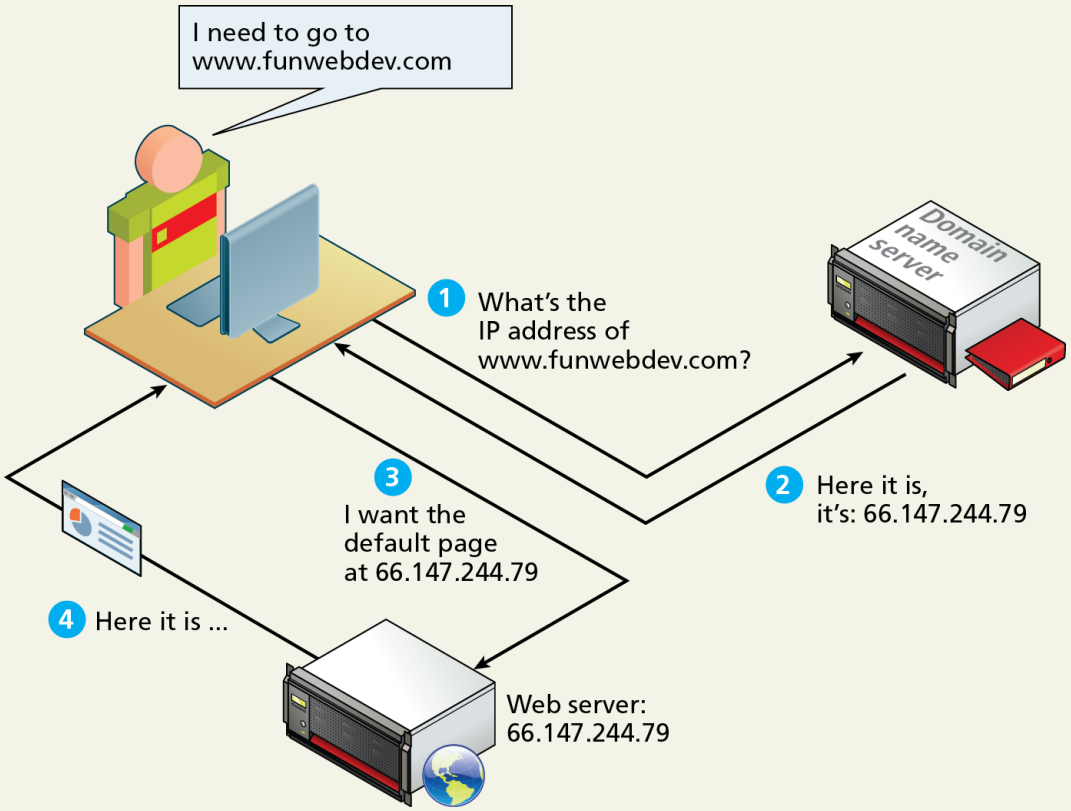
DOMAIN NAME SYSTEM (DNS)

Domain Name System

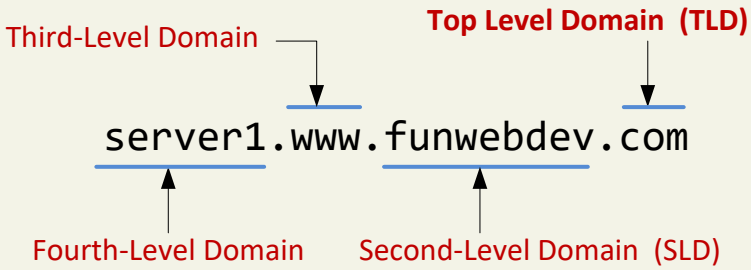
Why do we need it?

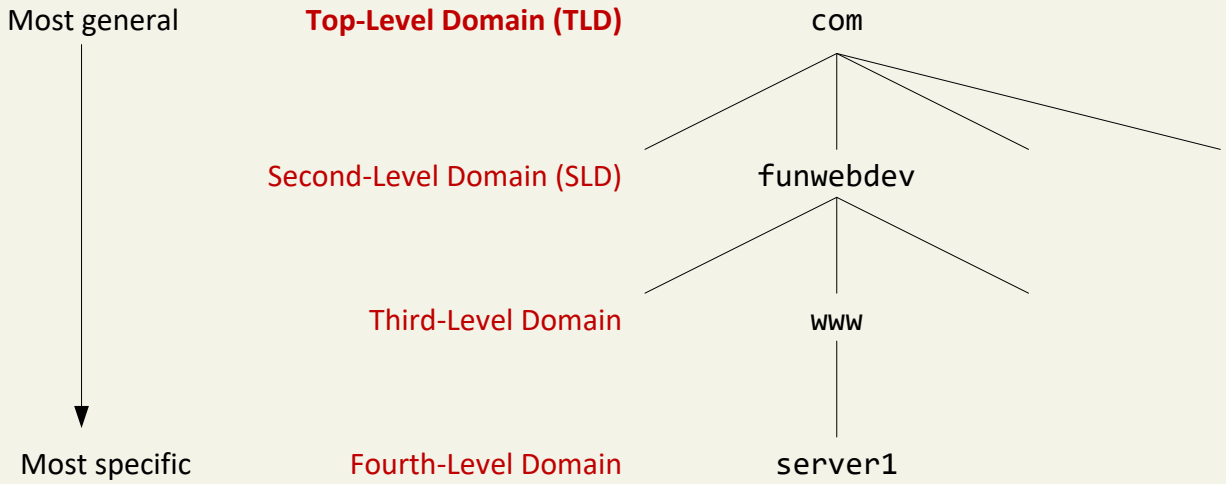
As elegant as IP addresses may be, human beings do not enjoy having to recall long strings of numbers.

Instead of IP addresses, we use the **Domain Name System** (**DNS**)



Domain Levels





Types of TLDs

- **Generic top-level domain** (gTLD)
  - Unrestricted. TLDs include .com, .net, .org, and .info.
  - Sponsored. TLDs including .gov, .mil, .edu, and others.
  - New TLDs.
- **Country code top-level domain** (ccTLD)
  - TLDs include .us , .ca , .uk , and .au.
  - Internationalized Domain Names

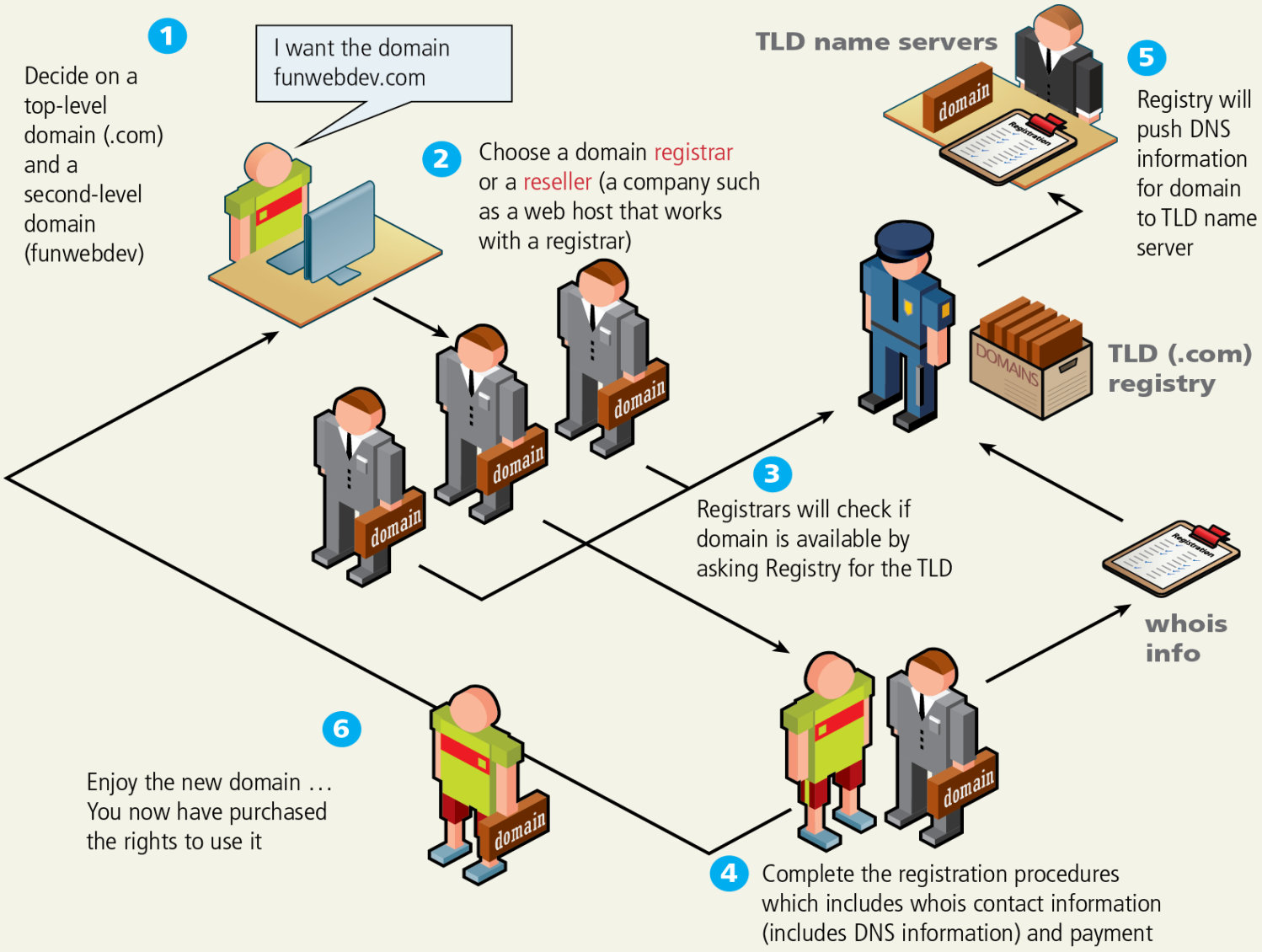
Name Registration

How are domain names assigned?

Special organizations or companies called **domain name registrars** manage the registration of domain names.

These domain name registrars are given permission to do so by the appropriate generic top-level domain (gTLD) registry and/or a country code top-level domain (ccTLD) registry.

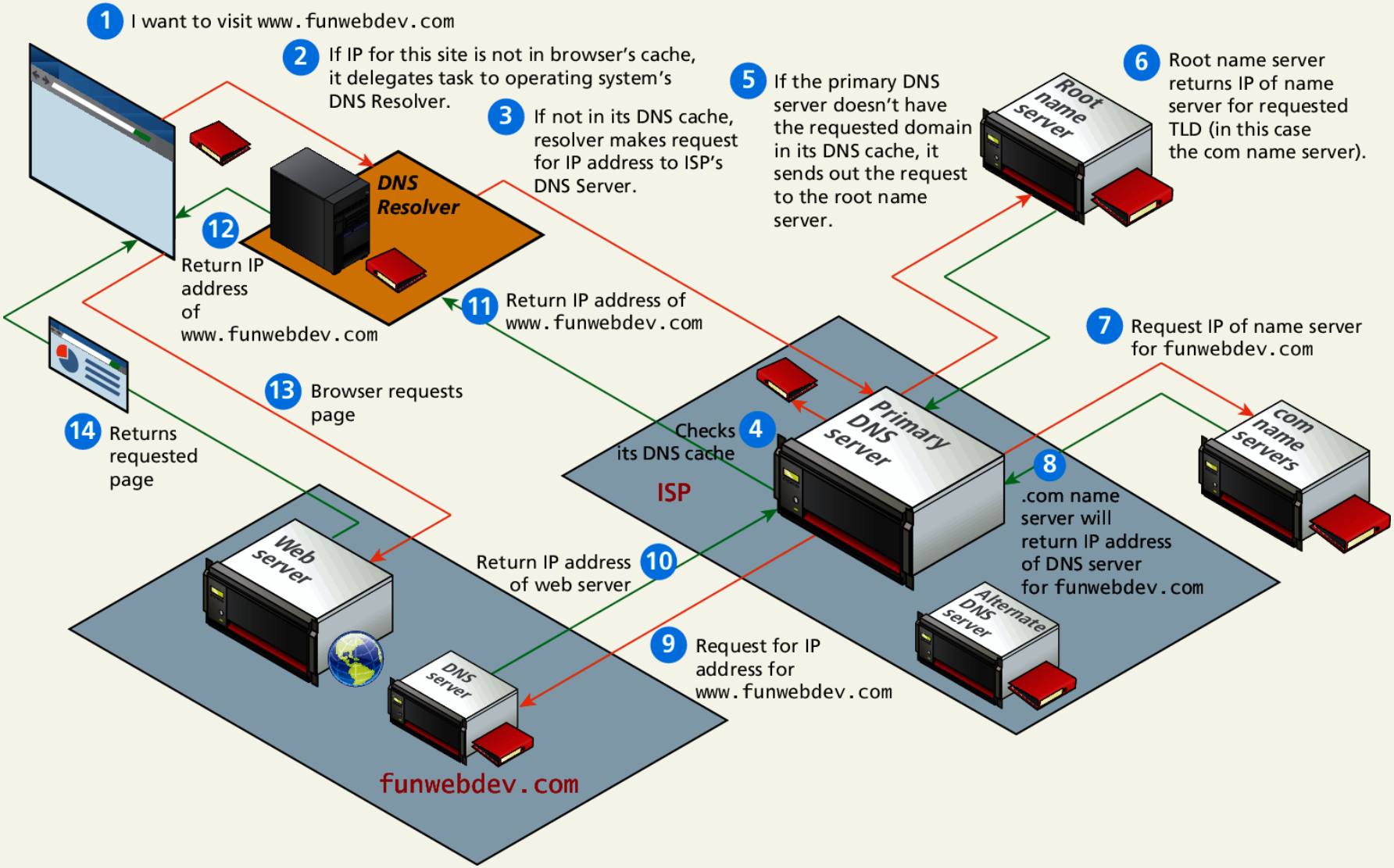
Domain name registration process



DNS Address Resolution

- While domain names are certainly an easier way for users to reference a web site, eventually, your browser needs to know the IP address of the web site in order to request any resources from it.
- The Domain Name System provides a mechanism for software to discover this numeric IP address.
- This process is referred to here as **address resolution**.

Domain name address resolution process



Section 6 of 8

UNIFORM RESOURCE LOCATORS (URL)

URL Components

In order to allow clients to request particular resources from the server, a naming mechanism is required so that the client knows how to ask the server for the file.

For the web that naming mechanism is the **Uniform Resource Locator** (**URL**).

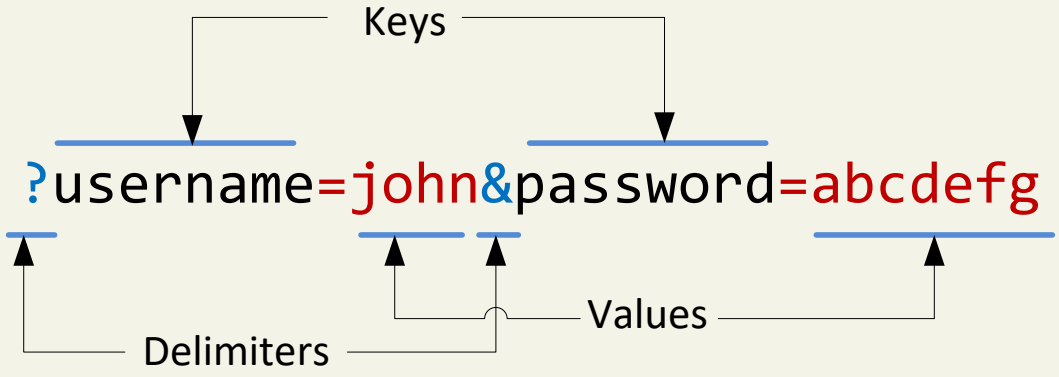


Query String

Query strings will be covered in depth when we learn more about HTML forms and server-side programming.

They are the way of passing information such as user form input from the client to the server.

In URL's they are encoded as key-value pairs delimited by "**&**" symbols and preceded by the "**?**" symbol.



Uniform Resource Locators

We mentioned several application layer protocols on top of the TCP/IP stack: FTP, SSH, HTTP, POP, IMAP, DNS, …

Requesting

- **ftp**://example.com/abc.txt → sends out an FTP request on port 21, while
- **http**://example.com/abc.txt → transmits an HTTP request on port 80.

Since the DNS system is case insensitive, the domain part of the URL is case insensitive

Alternatively, an IP address can be used for the domain

The optional port attribute allows us to specify connections to ports other than the defaults

Add a colon after the domain, then specify an integer port number.

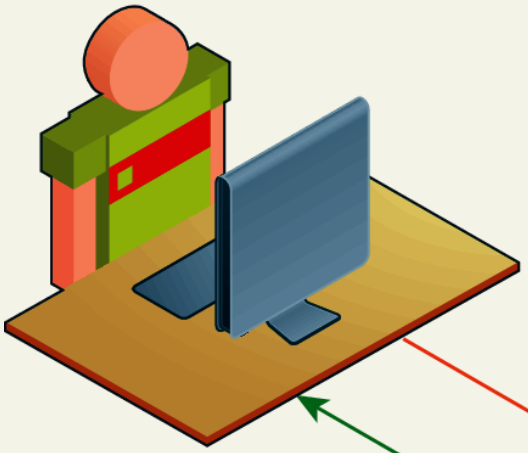
Section 7 of 8

HYPERTEXT TRANSFER PROTOCOL (HTTP)

HTTP

- The HTTP protocol establishes a TCP connection on port 80 (by default).
- Remember: ports for multiplexing and demultiplexing
- 80: **well-known** port
- The server waits for the request, and then responds with a response code, headers and an optional message (which can include file content).

HTTP



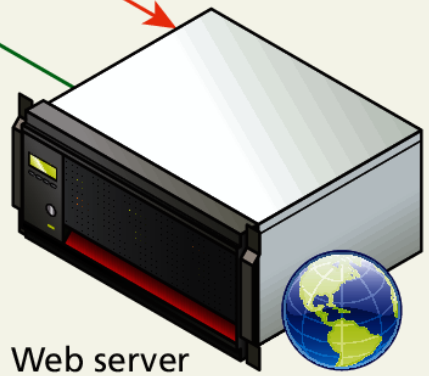
Request

```
GET /index.html HTTP/1.1
Host: example.com
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64;
rv:15.0) Gecko/20100101 Firefox/15.0.1
Accept: text/html,application/xhtml+xml
Accept-Language: en-us,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Cache-Control: max-age=0
```

Response

```
HTTP/1.1 200 OK
Date: Mon, 22 Oct 2012 02:43:49 GMT
Server: Apache
Vary: Accept-Encoding
Content-Encoding: gzip
Content-Length: 4538
Connection: close
Content-Type: text/html; charset=UTF-8

<html>
<head> ...
```

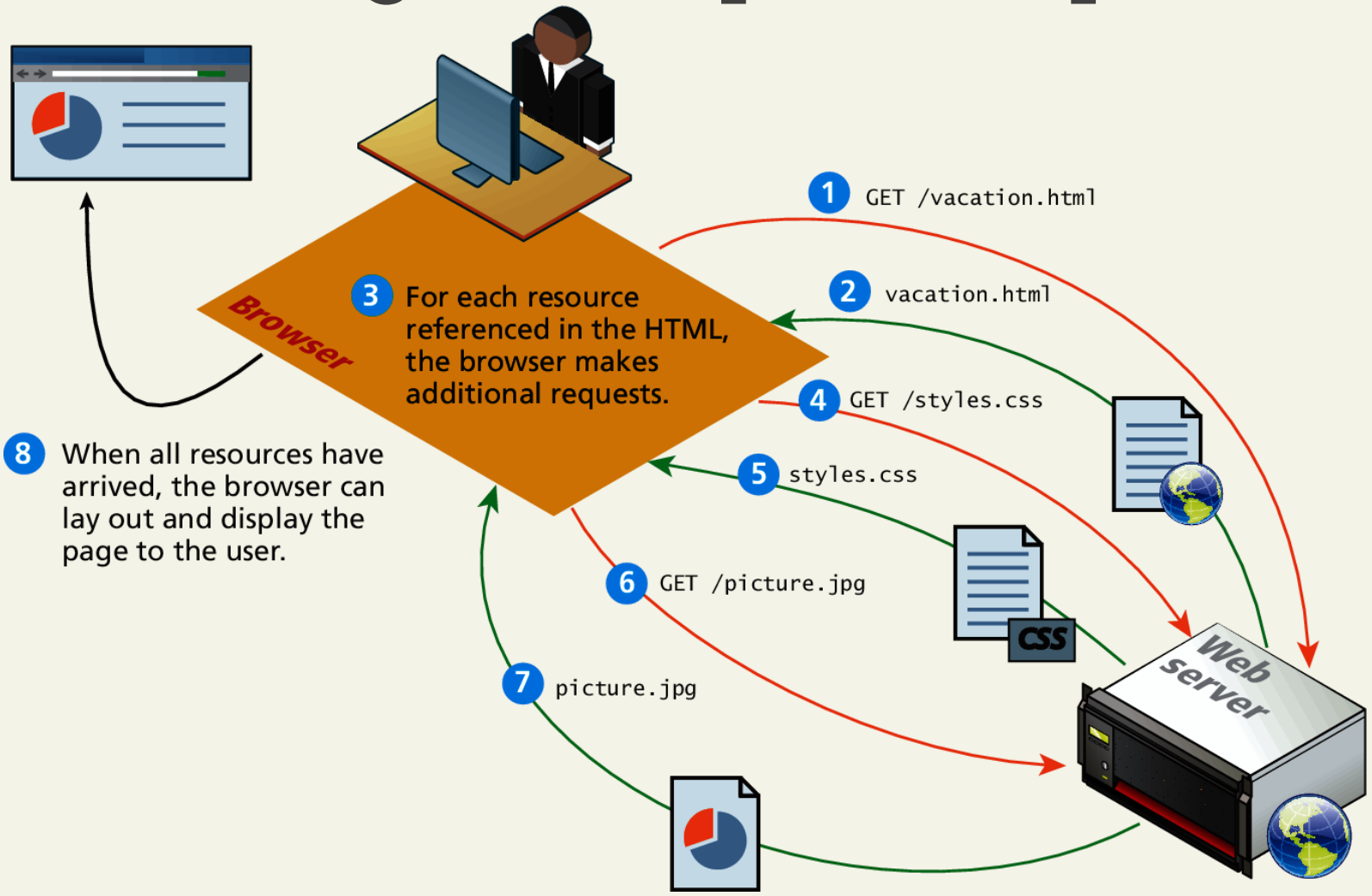


Web server

Web Requests

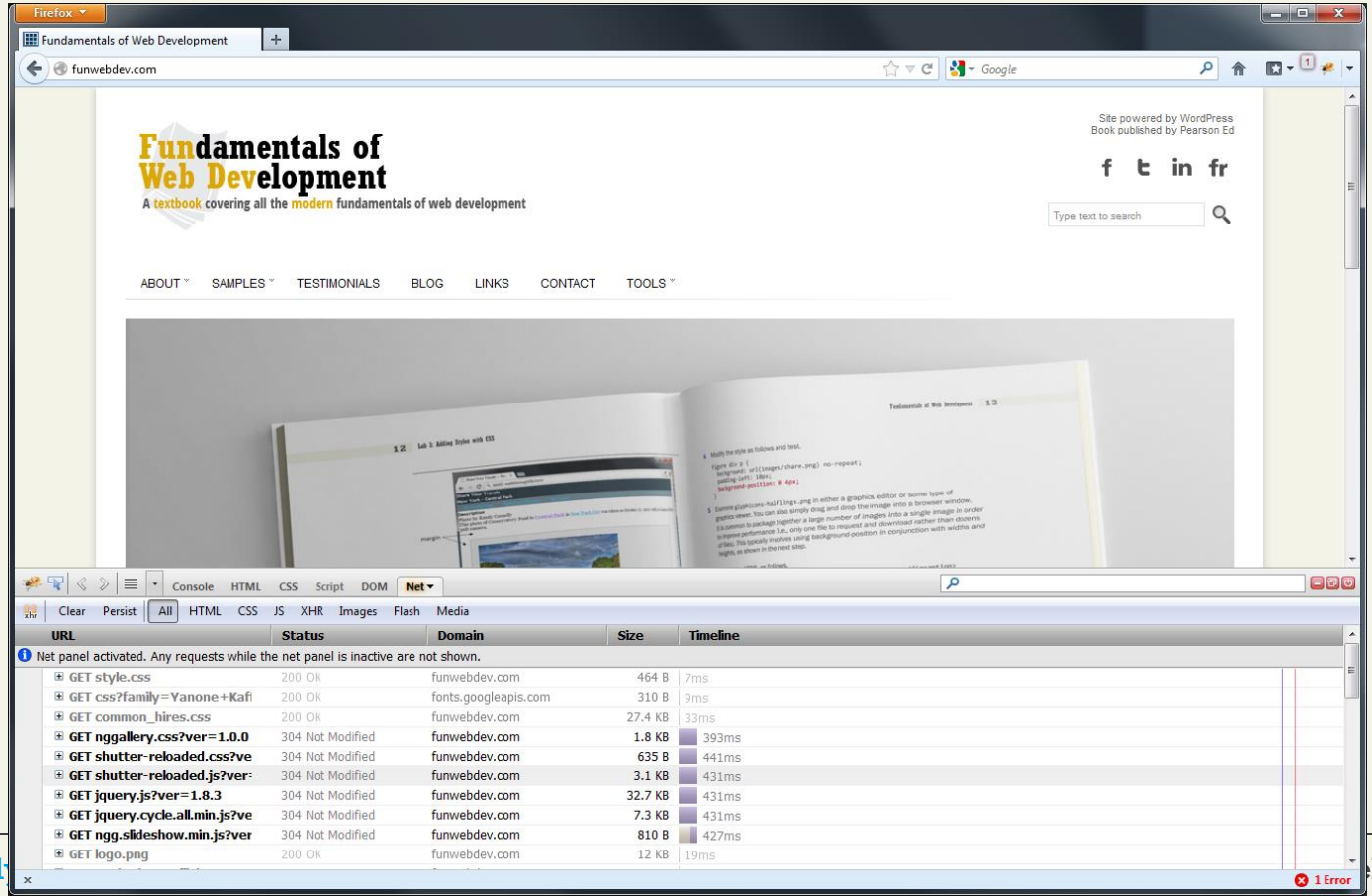
- While we as web users might be tempted to think of an entire page being returned in a single HTTP response, this is not in fact what happens.
- In reality the experience of seeing a single web page is facilitated by the client's browser which requests the initial HTML page, then parses the returned HTML to find all the resources referenced from within it, like images, style sheets and scripts.
- Only when all the files have been retrieved is the page fully loaded for the user

Browser parsing HTML and making subsequent requests



Browser Tools for HTTP

Modern browsers provide the developer with tools that can help us understand the HTTP traffic for a given page.



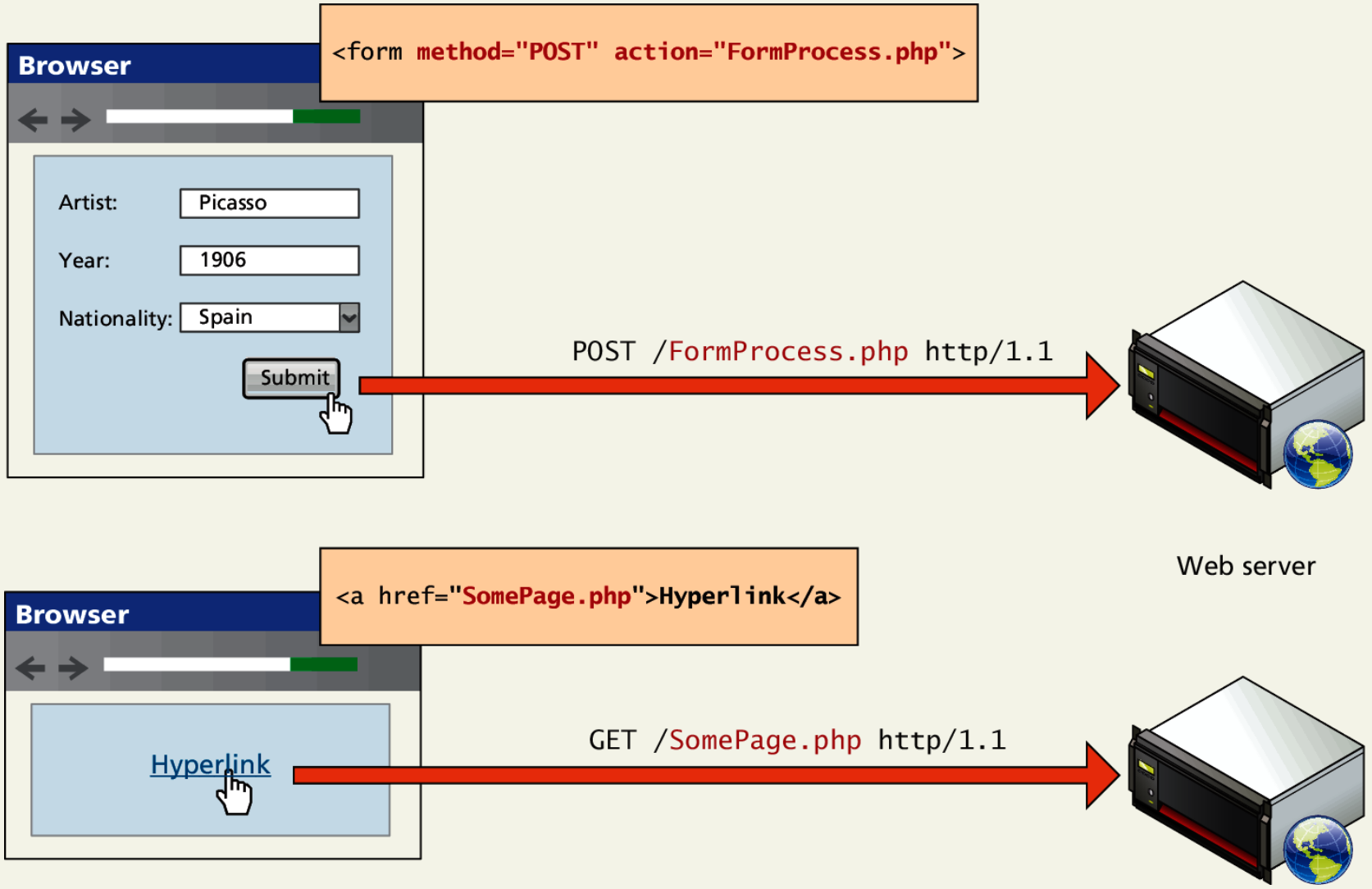
HTTP Request Methods

The HTTP protocol defines several different types of requests, each with a different intent and characteristics.

The most common requests are the **GET** and **POST** request, along with the HEAD request.

Other requests, such as PUT, DELETE, CONNECT, TRACE and OPTIONS are seldom used, and are not covered here.

GET versus POST requests



Hypertext Transfer Protocol

Response Codes

- 2## codes are for successful responses,
- 3## are for redirection-related responses,
- 4## codes are **client** errors,
- 5## codes are **server** errors.

Hypertext Transfer Protocol

(Some) Response Codes

200: OK

301: Moved Permanently

304: Not Modified

307: Temporary redirect

400: Bad Request

401: Unauthorized

404: Not found

414: Request URI too long

500: Internal server error

Section 8 of 8

WEB SERVERS

Web Servers

A **web server** is, at a fundamental level, nothing more than a computer that responds to HTTP requests.

Regardless of the physical characteristics of the server, one must choose an application stack to run a website.

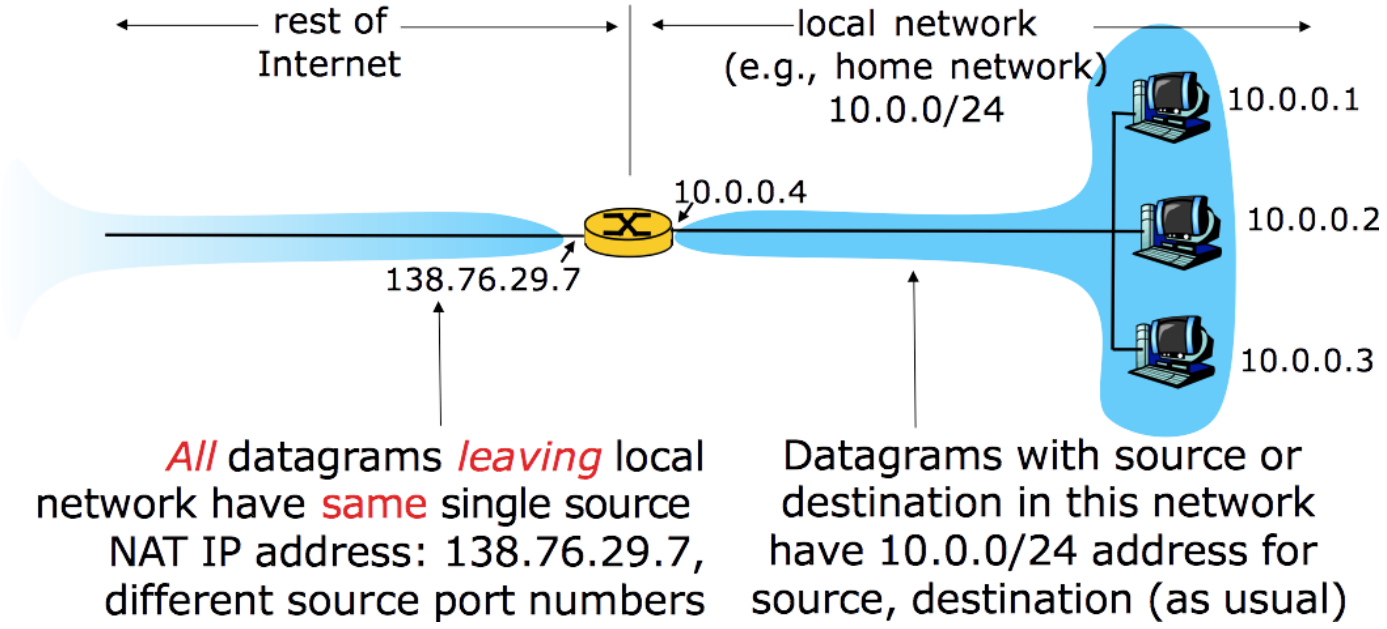
This stack will include an operating system, web server software, a database and a scripting language to process dynamic requests.

LAMP software stack, which refers to the Linux operating system, Apache web server, MySQL database, and PHP scripting language

WISA software stack, which refers to Windows operating system, IIS web server, SQL Server database, and the ASP.NET server-side development technologies.

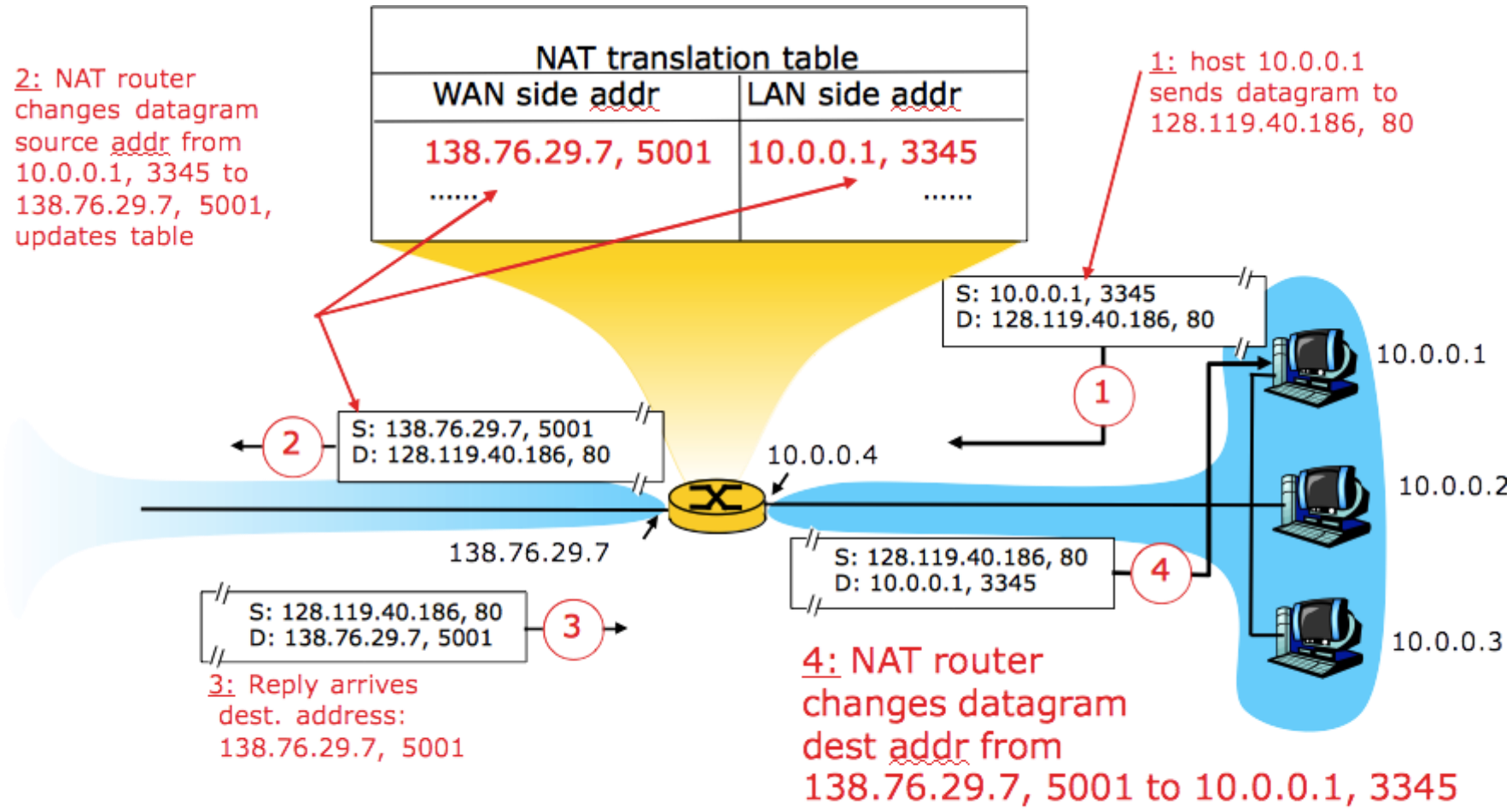
NAT: Network Address Translation

- Local network uses just one IP address as far as outside world is concerned



NAT: Network Address Translation

- Implementation: NAT router must:
  - outgoing datagrams: replace (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
    - . . . remote clients/servers will respond using (NAT IP address, new port #) as destination addr.
  - remember: in NAT translation table, every (source IP address, port #) to (NAT IP address, new port #) translation pair
  - incoming datagrams: replace (NAT IP address, new port #) in dest fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table


NAT translation table
WAN side addr
LAN side addr
138.76.29.7, 5001
.....
10.0.0.1, 3345
.....
2: NAT router changes datagram source addr from 10.0.0.1, 3345 to 138.76.29.7, 5001, updates table
1: host 10.0.0.1 sends datagram to 128.119.40.186, 80
S: 10.0.0.1, 3345
D: 128.119.40.186, 80
1
10.0.0.1
10.0.0.2
10.0.0.3
10.0.0.4
2
S: 138.76.29.7, 5001
D: 128.119.40.186, 80
138.76.29.7
S: 128.119.40.186, 80
D: 10.0.0.1, 3345
4
S: 128.119.40.186, 80
D: 138.76.29.7, 5001
3
3: Reply arrives dest. address: 138.76.29.7, 5001
4: NAT router changes datagram dest addr from 138.76.29.7, 5001 to 10.0.0.1, 3345